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Research

AI at 70: 14 lessons from a lifetime of boom and bust

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With deep dedication.

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Artificial intelligence marks 70 years since Dartmouth Summer Research Project

“Context is all you need”: clues from the past for what comes next



Seventy years ago today, the curtain came down on the summer workshop that gave birth to artificial intelligence. Dwight Eisenhower was in the White House, the Suez Crisis was gripping Britain and Elvis Presley topped the US charts with “Don’t Be Cruel” and “Hound Dog”.

The Dartmouth Summer Research project brought together a dozen or so academics convinced that every aspect of intelligence could in principle be described precisely enough for a machine to simulate it. Emboldened by the techno-optimism of the age, John McCarthy, who coined the name AI, wrote that “we think that a significant advance can be made... if a carefully selected group of scientists work on it together for a summer.”

Since then, AI has lived a lifetime of summers and winters, punctuated by stunning breakthroughs and false dawns. As it is now the cornerstone of a new investment and valuation boom, the past 70 years are a treasure trove of clues for what to expect in the future.

To paraphrase the research paper that sparked the current boom in generative AI in 2017, “Attention is all you need”, you could say “context is all you need” to understand what is to come. The signals are there, from non-linear growth to infrastructure constraints and valuation booms and busts.

Some will say “this time is different”. Maybe. But the following charts give a flavour of what could happen if it isn’t. To quote the UK No. 1 single from 70 years ago, sung by Doris Day: “Whatever Will Be, Will Be (Que Sera, Sera)”.

14 lessons from a lifetime of boom and bust

- | | |
|---|--|
| 1. AI progress is non-linear | cheaper, and more at risk |
| 2. Progress exceeds Moore's law | 9. Tech revolutions take time to pay |
| 3. Today's leading tech may not be tomorrow's | 10. Consumers have adopted AI faster than businesses |
| 4. The pipeline matters | 11. The biggest valuation booms are often tech booms |
| 5. Cheaper AI will likely boost spending | 12. This time is different – and will need to be |
| 6. Software revolutions need hardware | 13. Early leads do not guarantee long-term winners |
| 7. Hardware is often a bottleneck | 14. Long-term trends tend to persist |
| 8. Globalisation has made tech | |

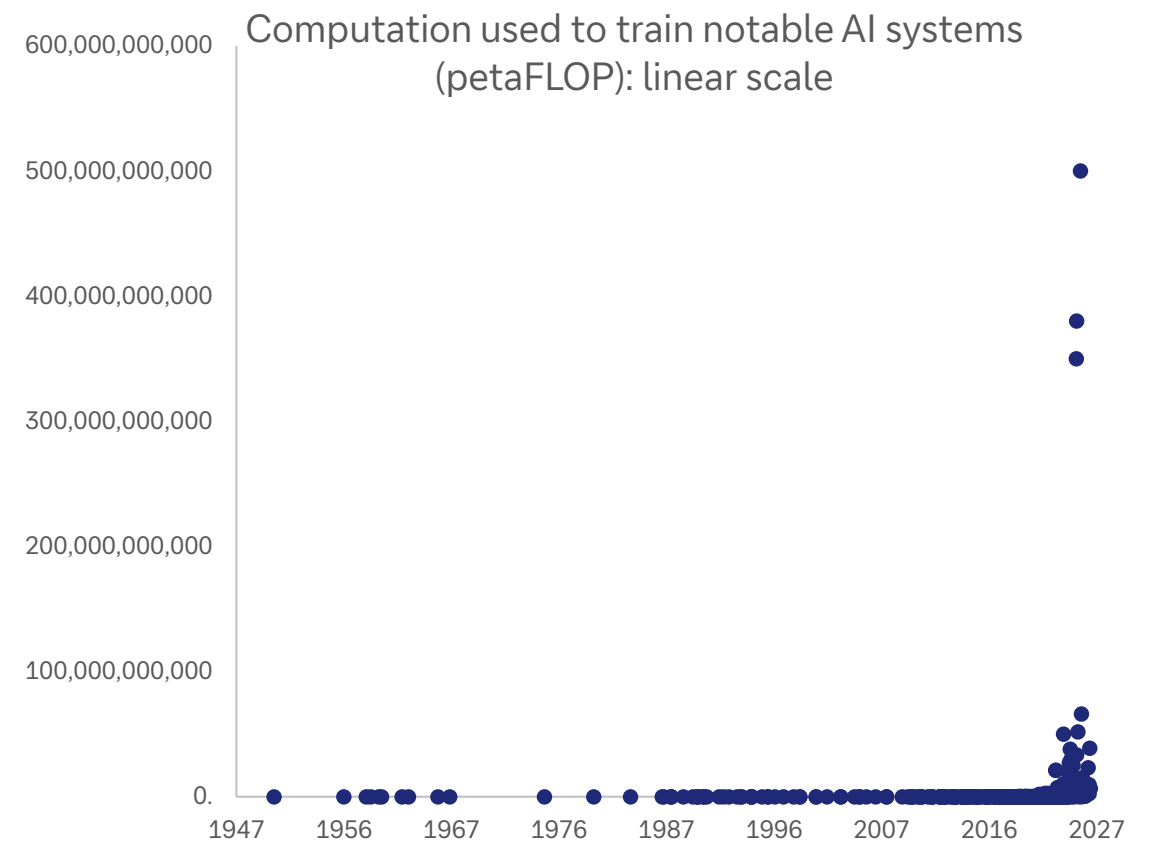
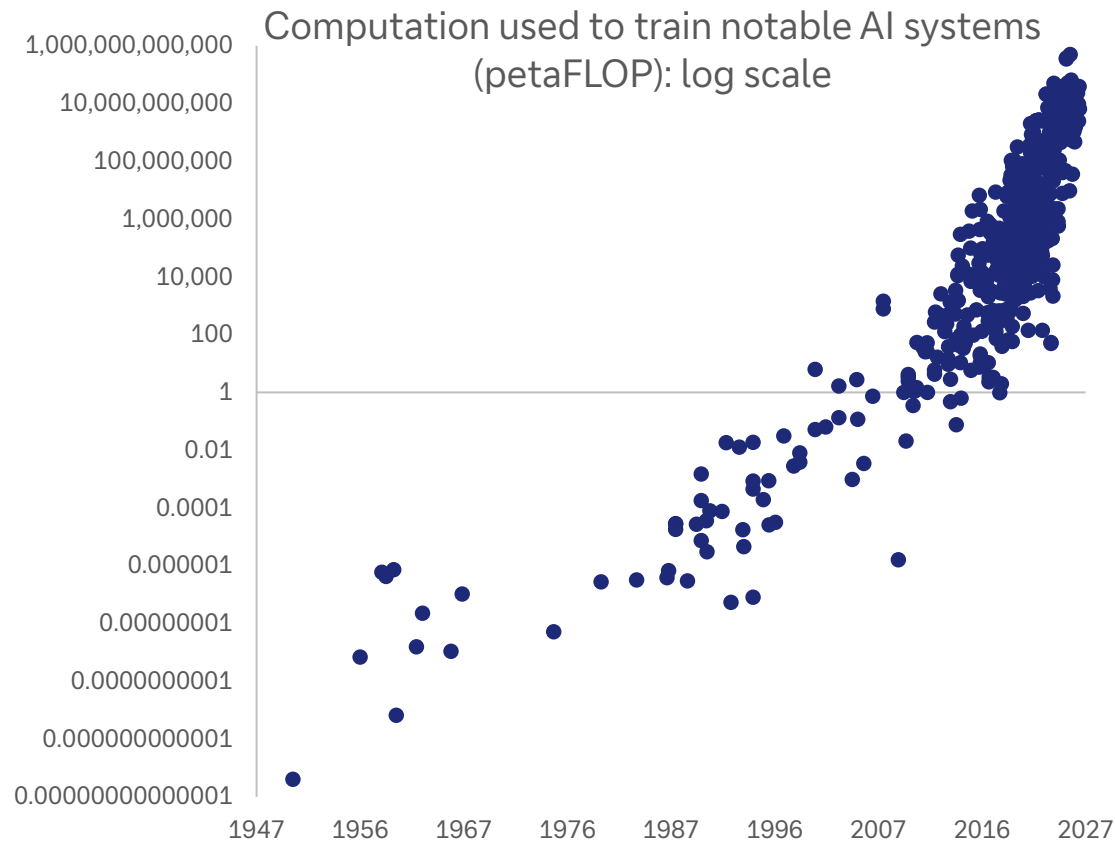
A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so pre-

1. AI progress is non-linear – and exponential growth is almost invariably underestimated

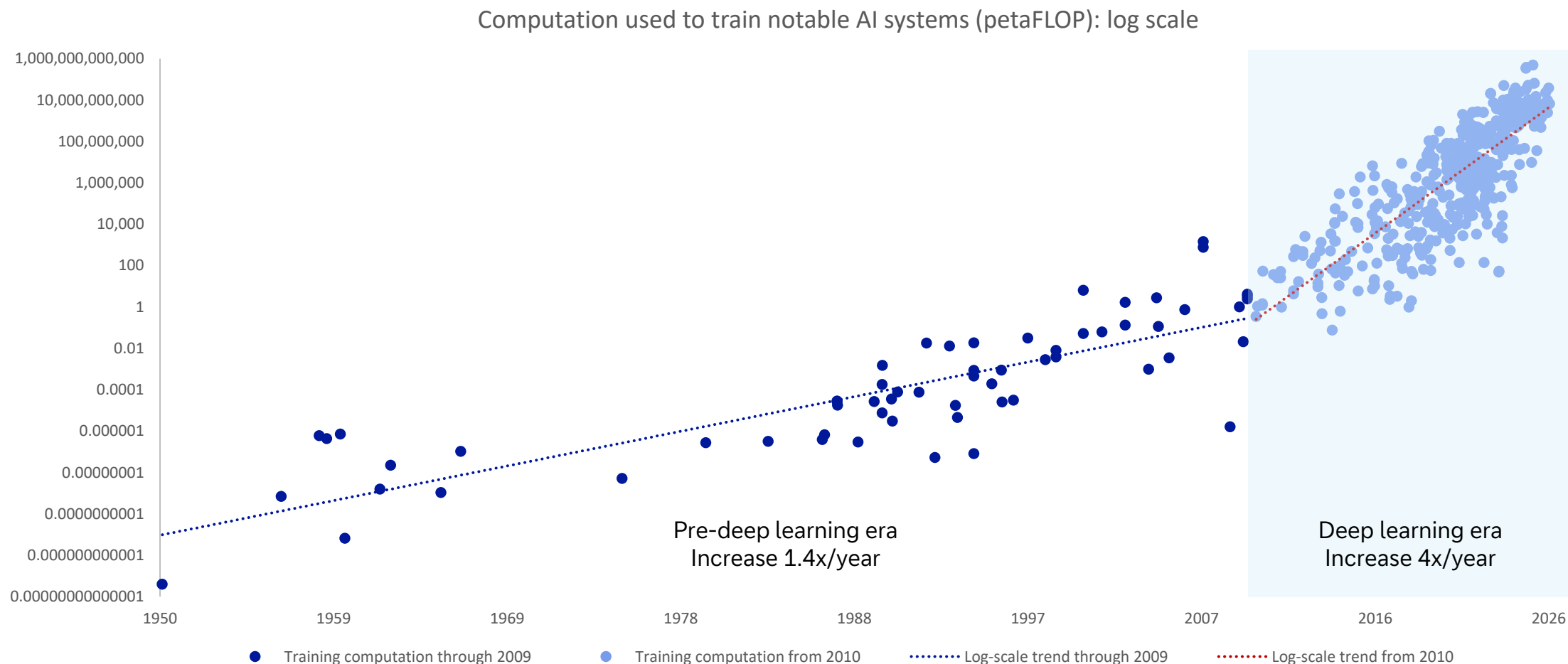
Compare what AI charts usually look like (on the left) vs what they actually represent (on the right)



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research. Note: One FLOP (floating-point operation) represents a single arithmetic operation using floating point numbers; one petaFLOP is one quadrillion FLOPs

2. The pace of progress exceeds Moore's law when many factors are improving at once

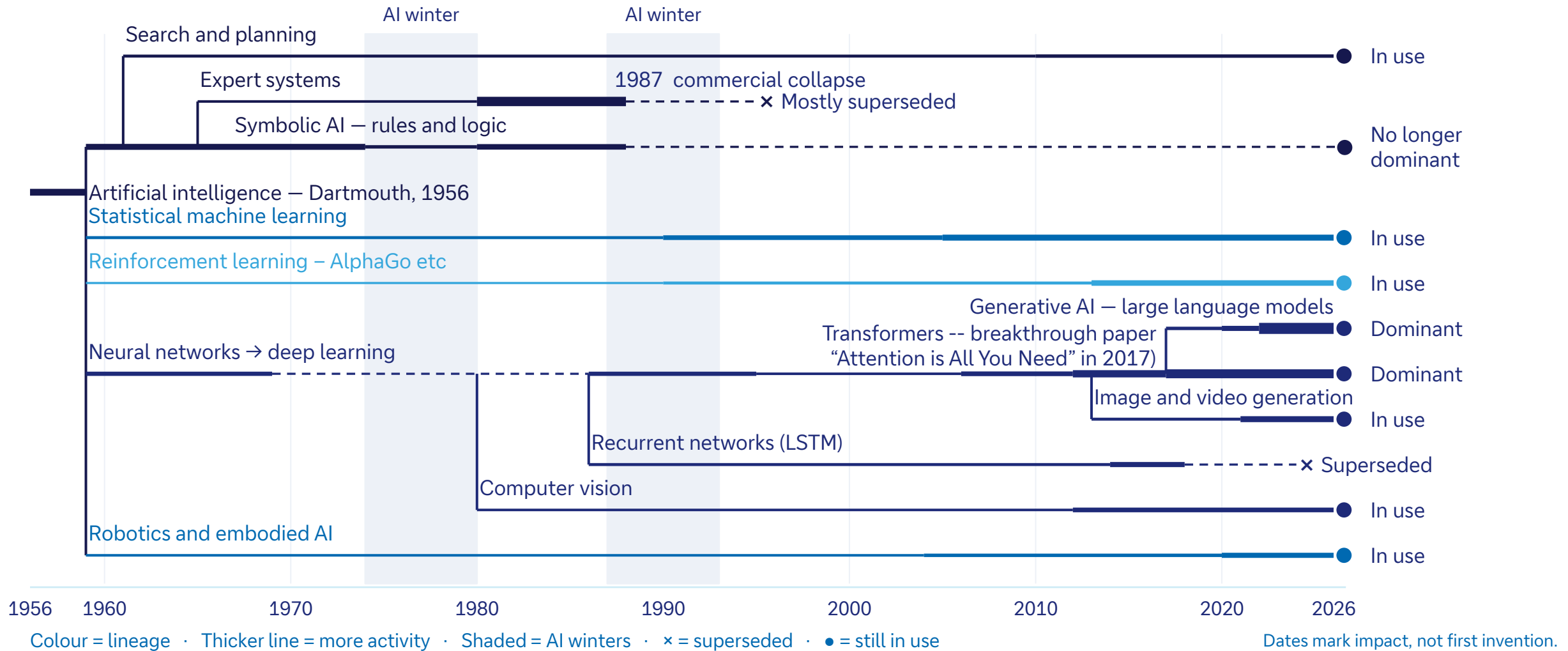
Conventional computing capability has doubled every 18-24 months but bigger and better systems, memory and algorithms have accelerated capabilities far beyond that



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

3. Today's leading tech may not be tomorrow's

AI has an ever-evolving family tree; large language models may give way to, eg, world models



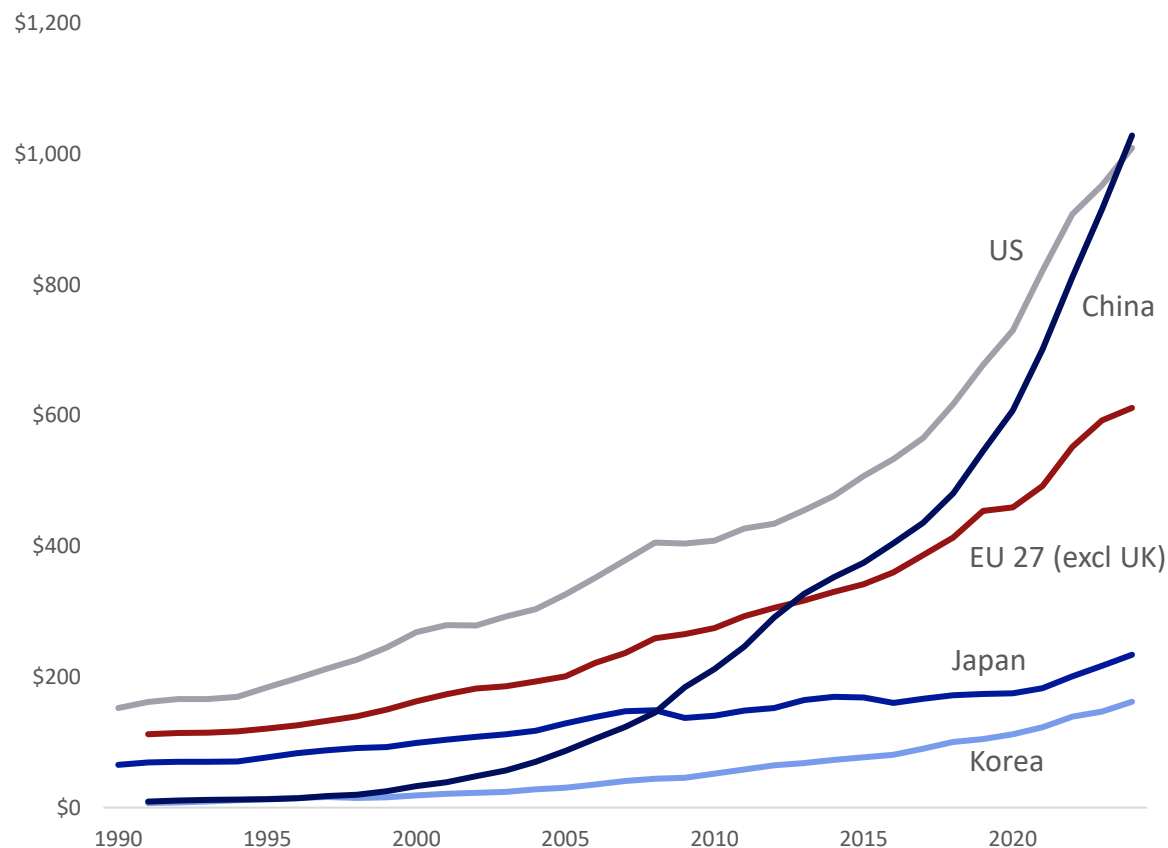
Source: Deutsche Bank Research

4. The pipeline matters: China's overnight success with DeepSeek was years in the making

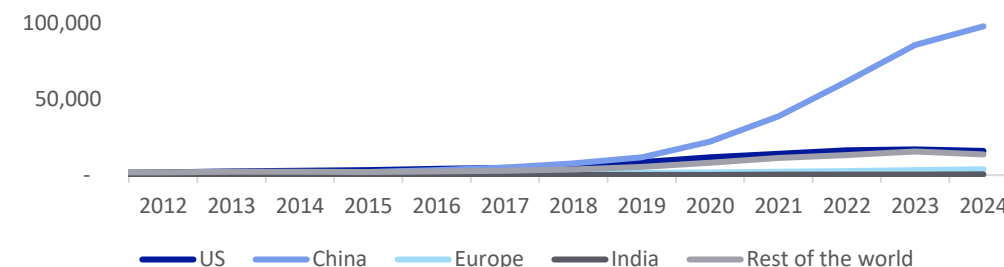
Success of Chinese models reflects a multi-year push in R&D, overtaking the US in 2024



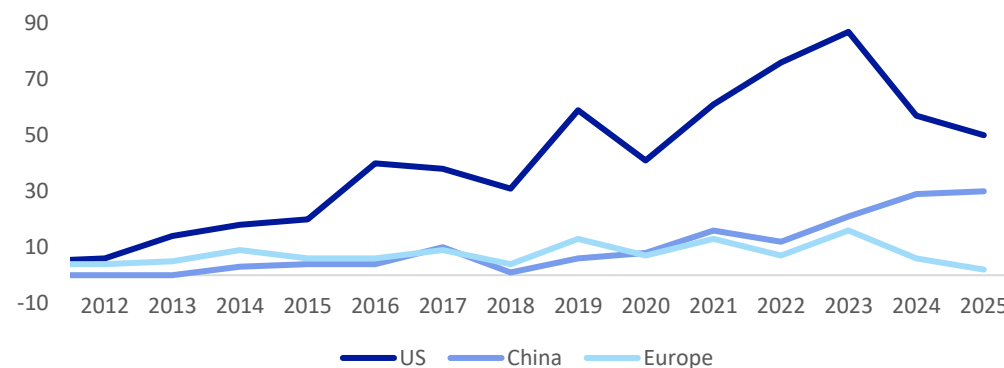
Gross domestic research and development spend by country,
PPP converted, current prices (USDbn)



Number of AI patents granted by select geographic
areas, 2012-2024



Number of notable AI models by geographic area
(2012-2025)



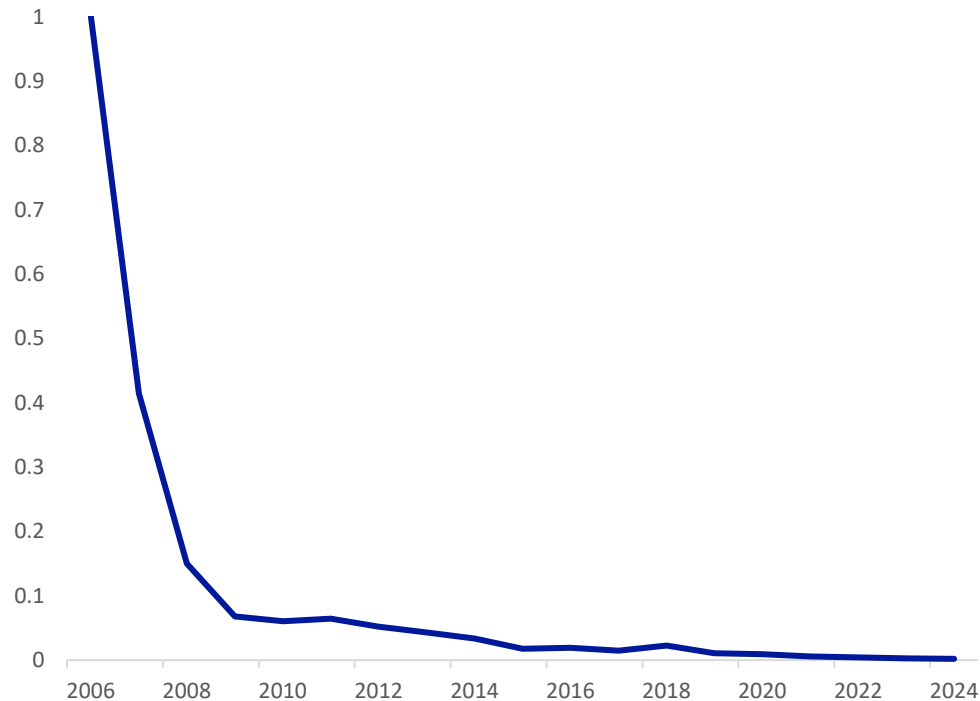
Source: OECD Data Explorer, Deutsche Bank Research

5. Making AI dramatically cheaper will likely increase, not decrease, spending

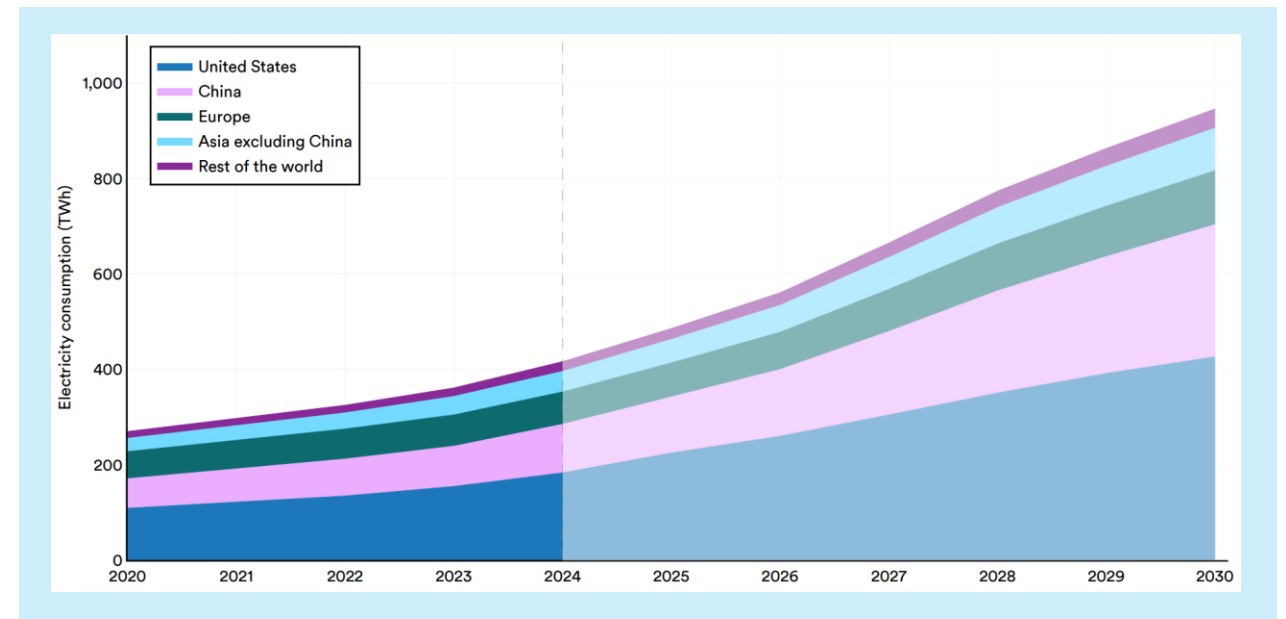
Jevons Paradox is alive: GPU computation cost has fallen by more than 99% since 2006, yet electricity consumption by data centres is set to double from 2024 to 2030 as demand surges



GPU chip computation cost index (2006=1)



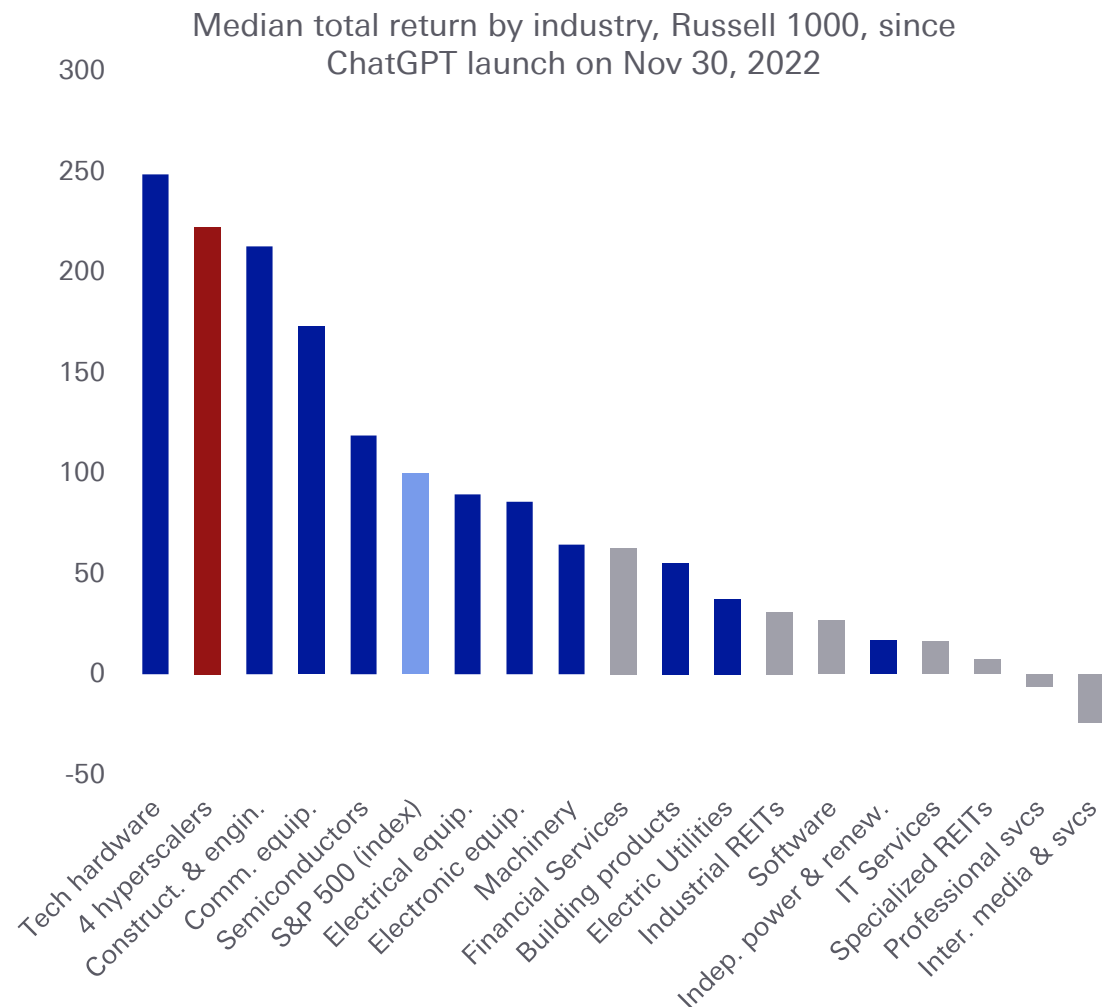
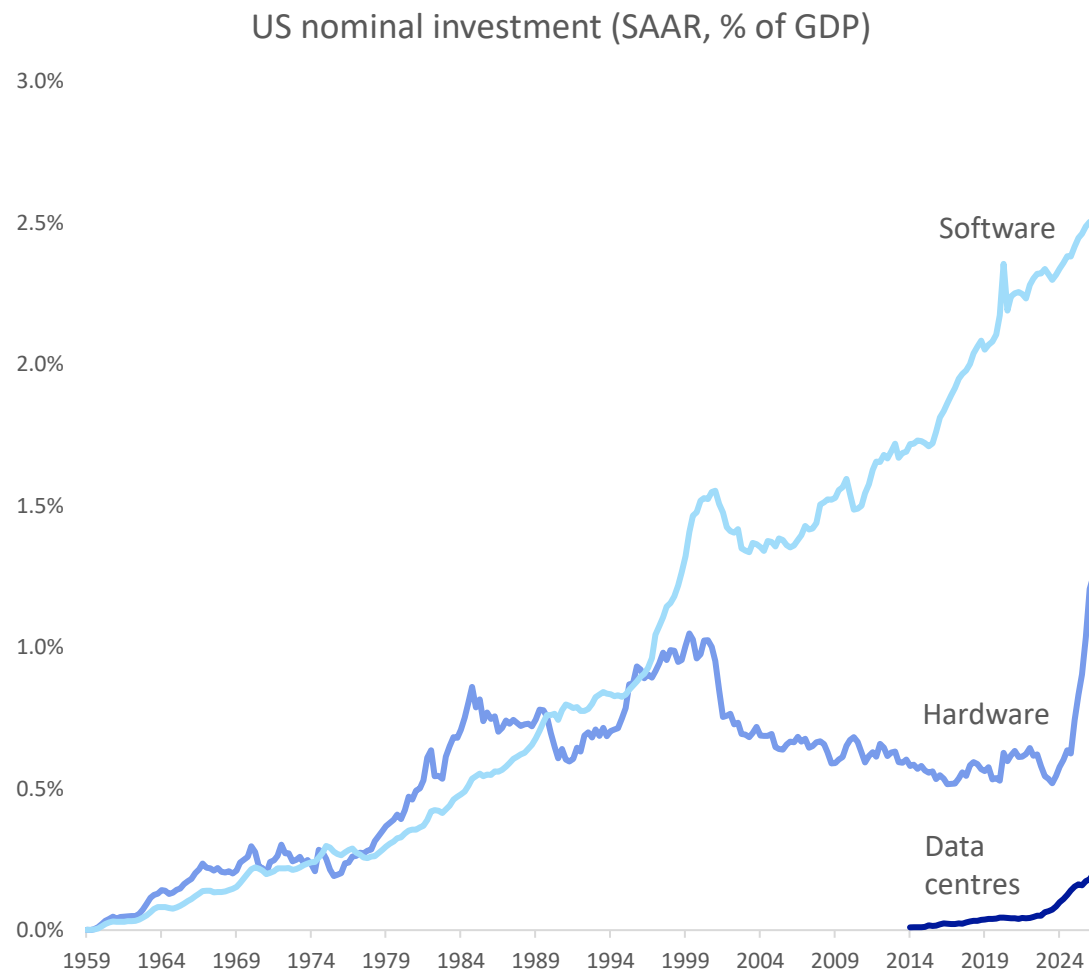
Data centre electricity consumption by region
2020-2030 (IEA projections)



Source: International Energy Agency (2025), Stanford HAI AI Index 2026, Deutsche Bank Research

6. Software revolutions depend on heavy hardware investment

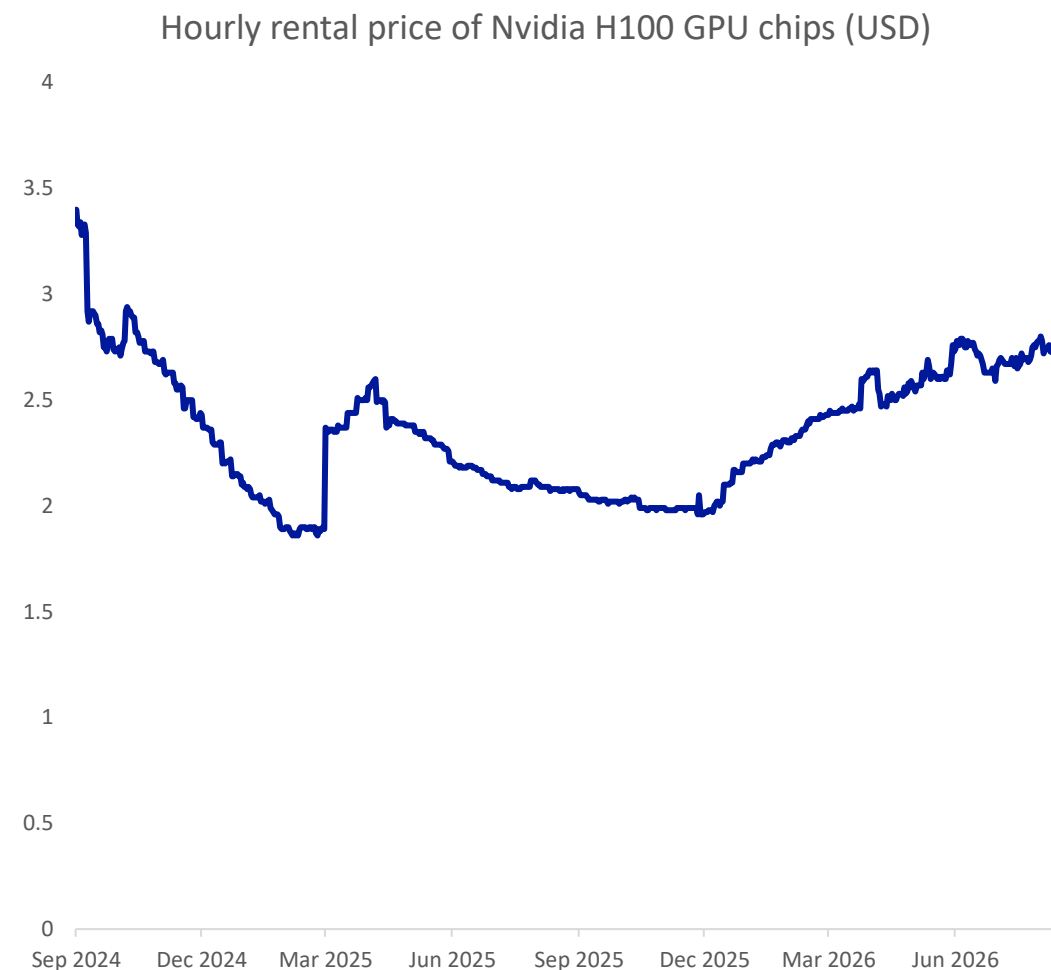
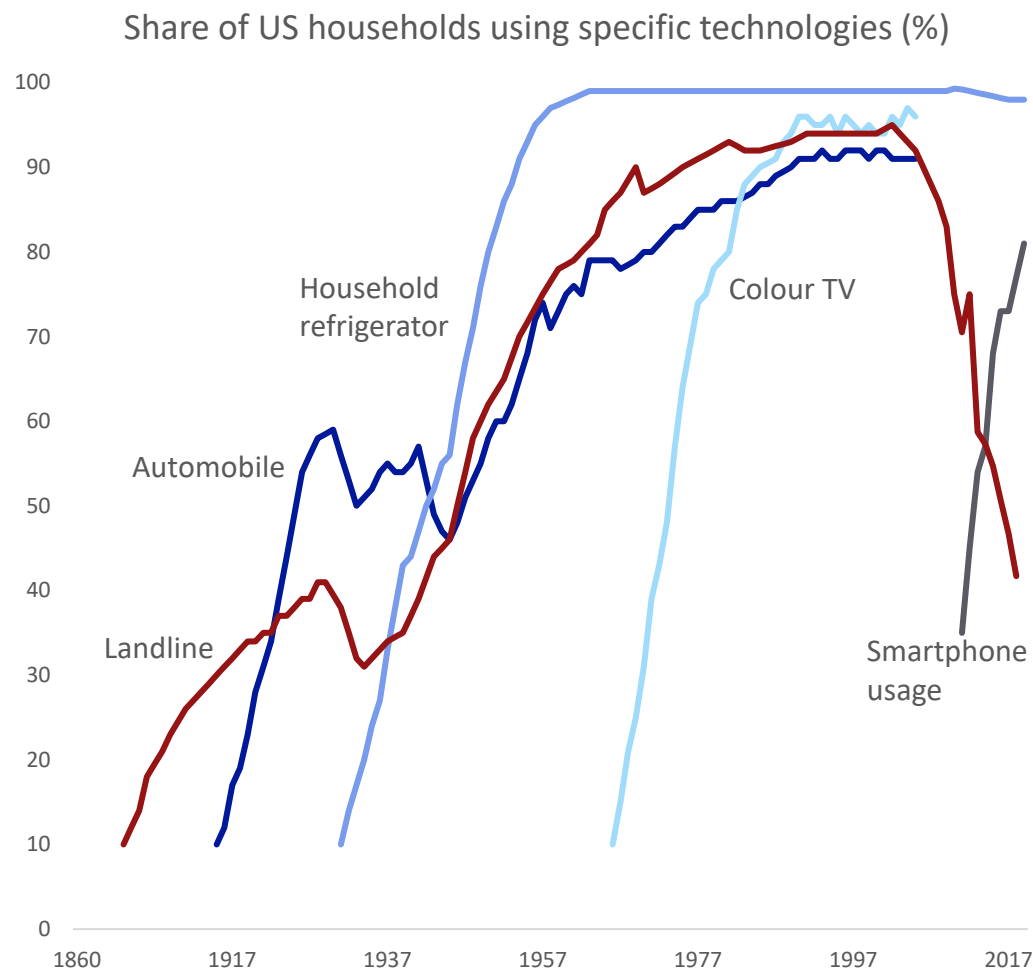
Hardware, construction, chip and equipment makers have led gains in the AI boom so far



Source: Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

7. Hardware is often a bottleneck – but for once the delay is with producers not consumers

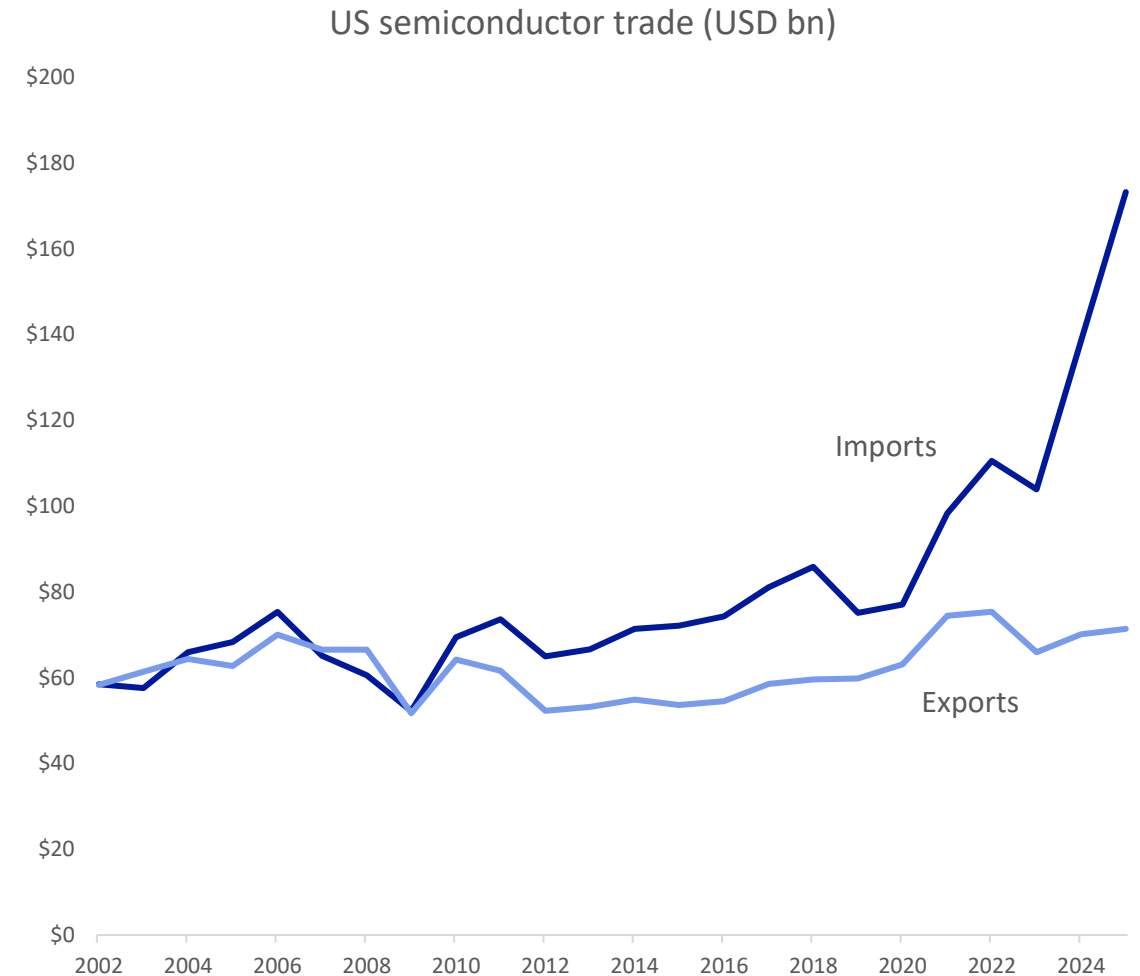
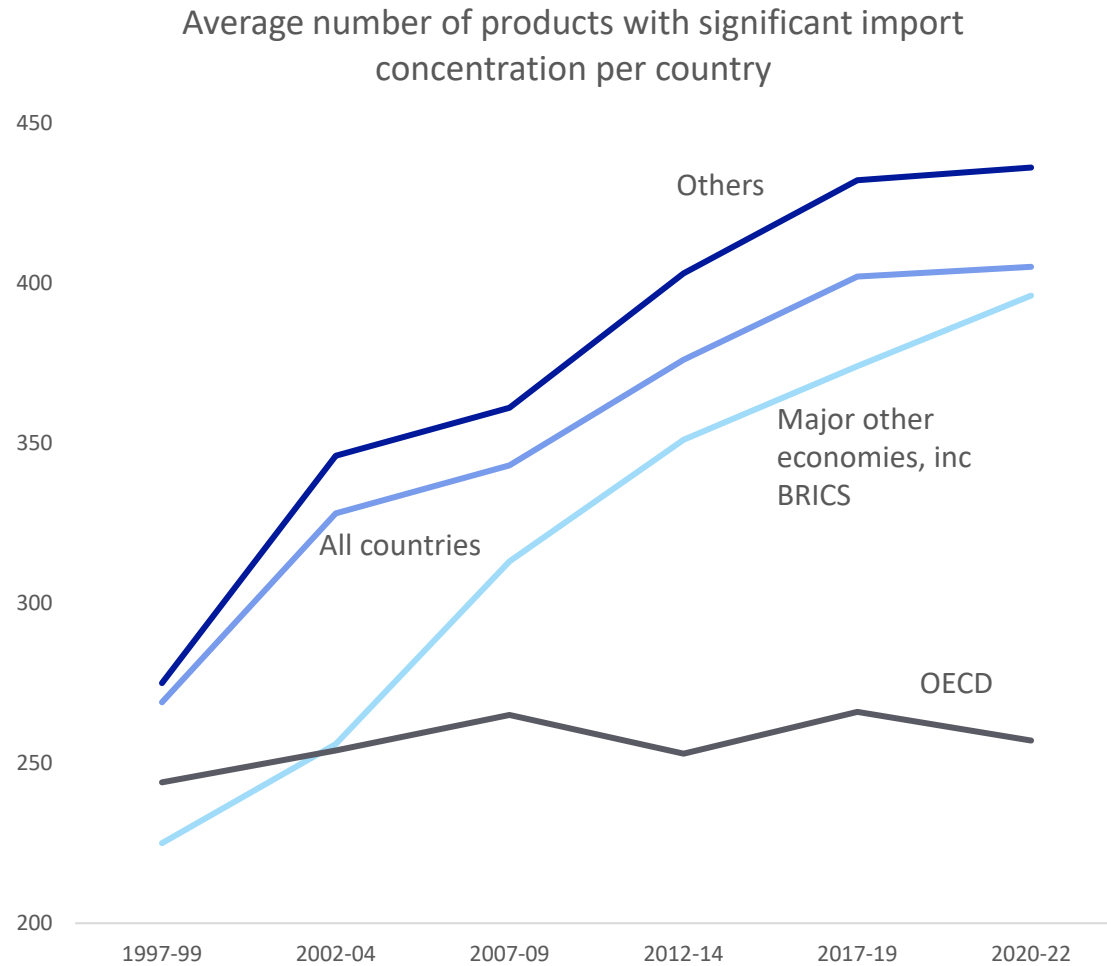
Past booms often relied on getting devices to consumers; now it's about getting scarce hardware online to serve them remotely



Source: Our World in Data, Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

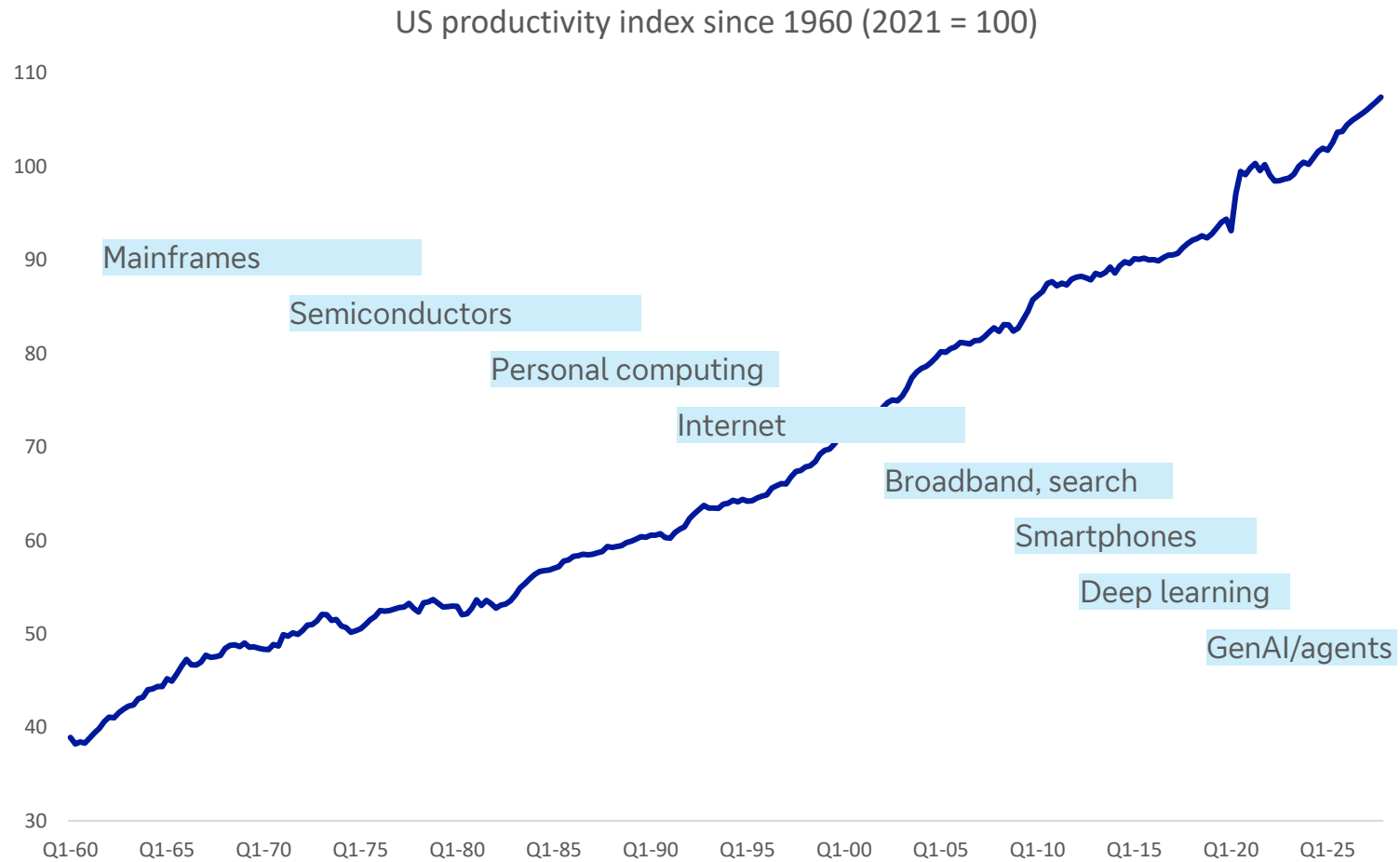
8. Globalisation made tech cheaper – but also increased supply chain risks

There has been a surge in US imports of semiconductors as the AI boom has taken off

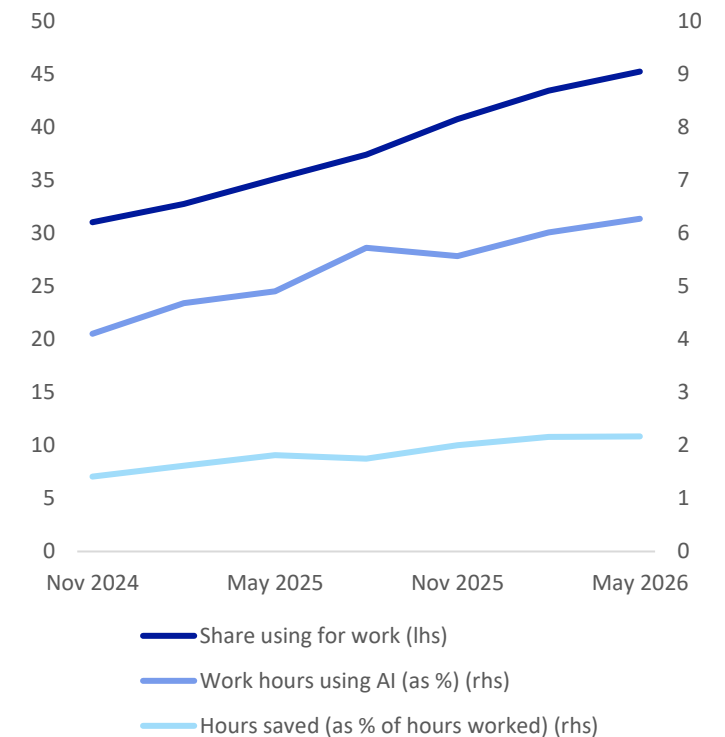


9. Tech revolutions take time to pay off, making them hard to see in long-term data

General purpose technologies have a productivity J-curve, generating costs before results



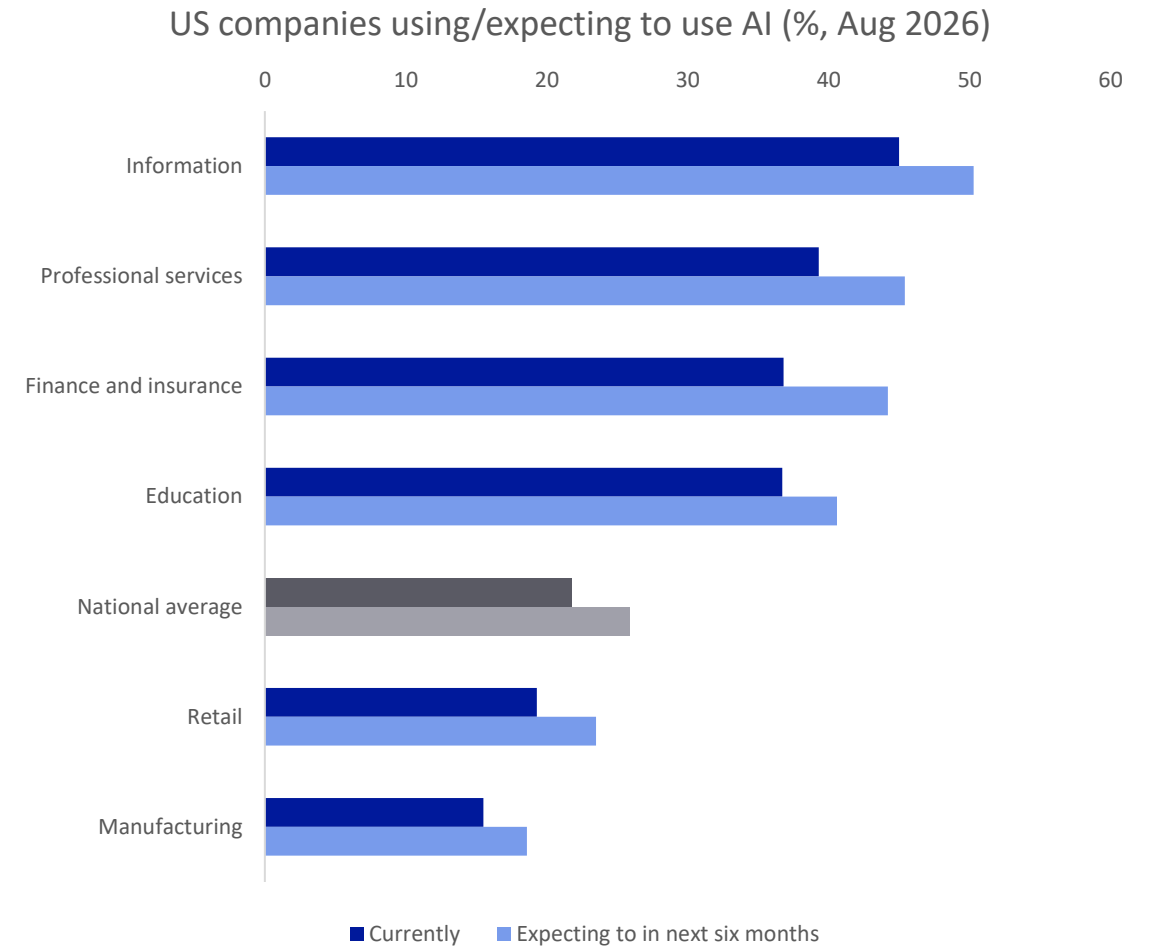
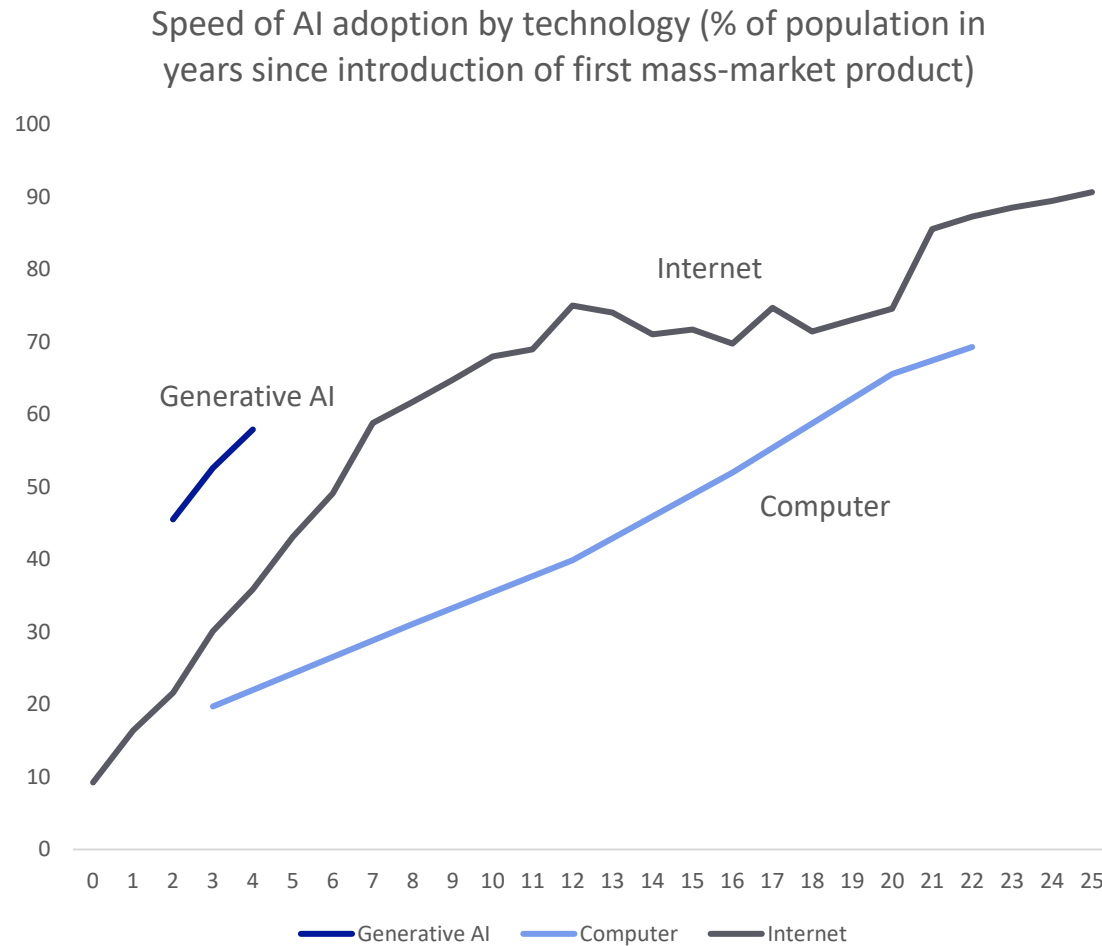
Fewer than half of US workers say they are using AI at work, with a mean of just 6% of time and saving 2% of hours



Source: OECD, GenAI Adoption Tracker, Deutsche Bank Research; blue bars in productivity chart give rough indication of notable tech eras

10. Consumers have adopted AI at a record pace but monetisable business use is slower

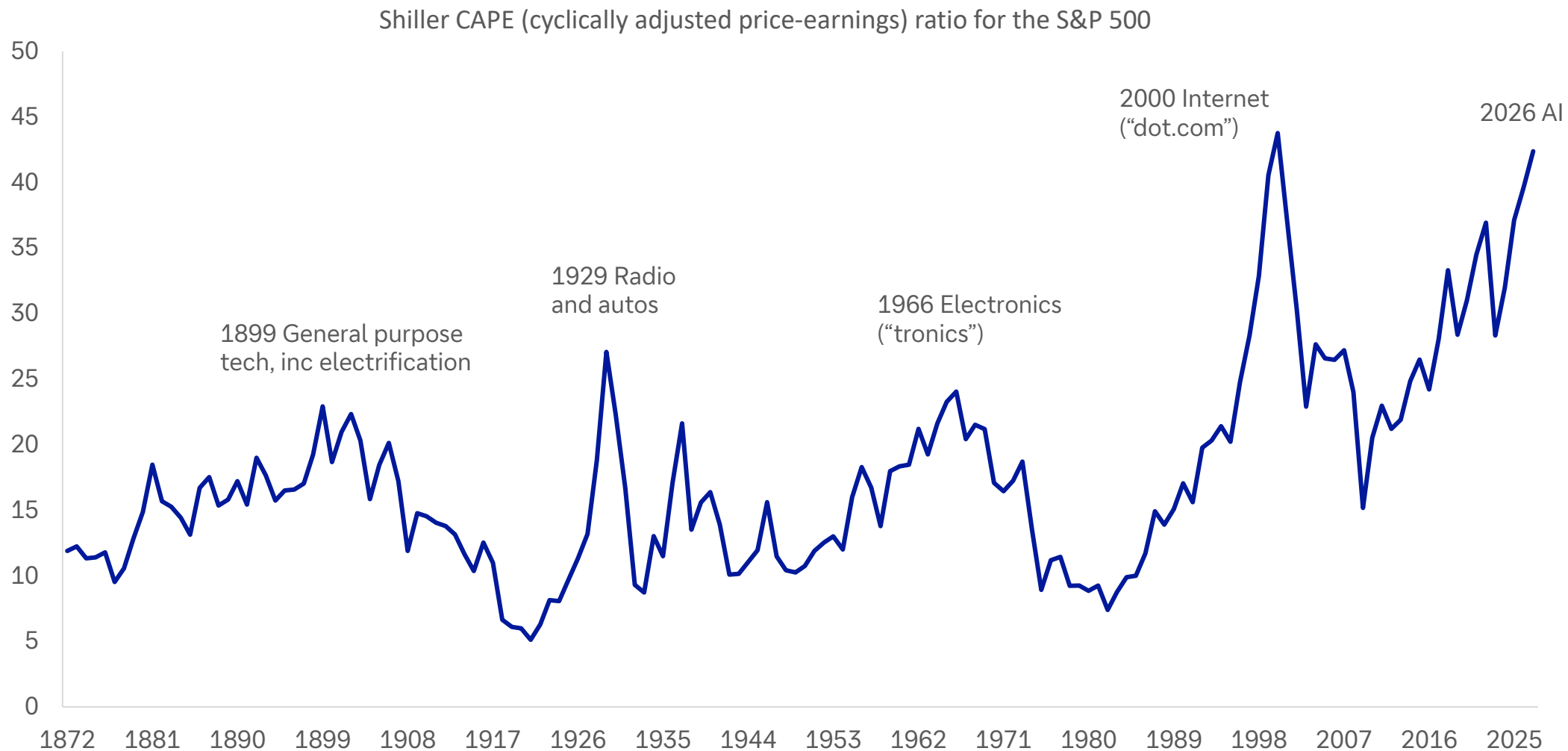
Downloading an app takes minutes; reorganising enterprises round new tech takes years



Source: GenAI Adoption Tracker, US Census Bureau, Deutsche Bank Research

11. The biggest valuation booms are often tech booms – like today

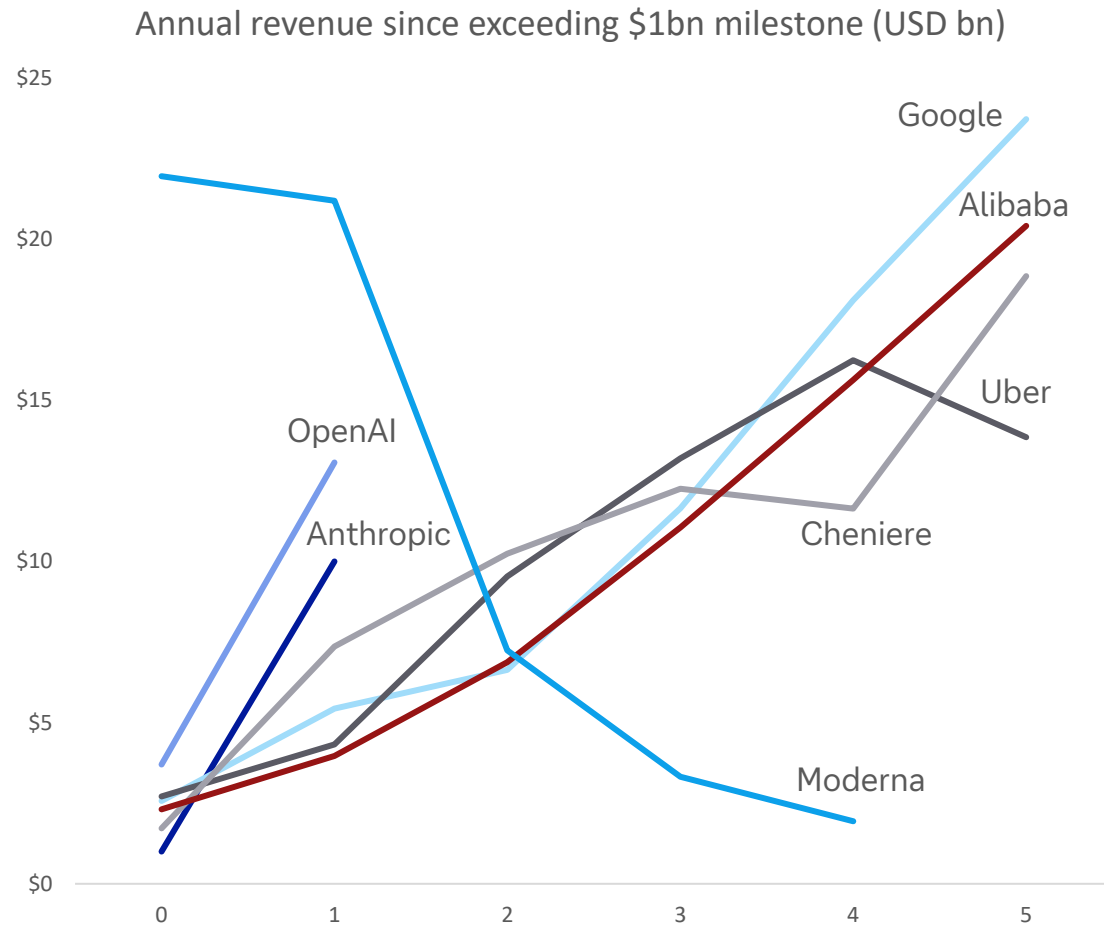
Peaks have mostly coincided with transformative technologies



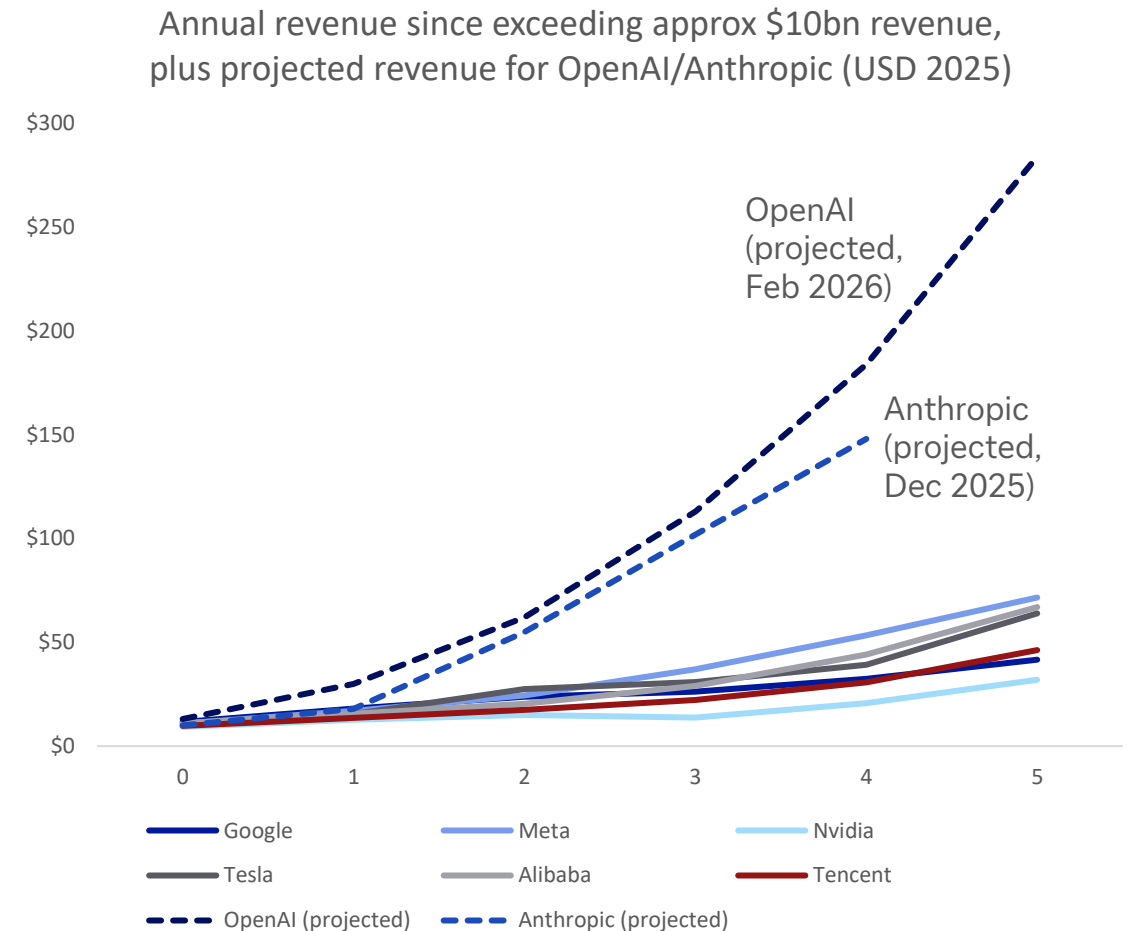
Source: Robert Shiller, Deutsche Bank Research

12. This time is different – but it will need to be even more different to meet projections

OpenAI and Anthropic are exceeding record revenue growth set by peers, at least for now



Source: Bloomberg Finance LP, media reports, Deutsche Bank Research. Note: Year 0 shown as first year revenue exceeded \$1bn (2025 USD). For OpenAI and Anthropic that was reportedly 2025. Moderna nominal revenue spiked from \$800m in 2020 to \$18.5bn in 2021, due to Covid vaccine. Cheniere revenue spiked with 2022 energy crisis.



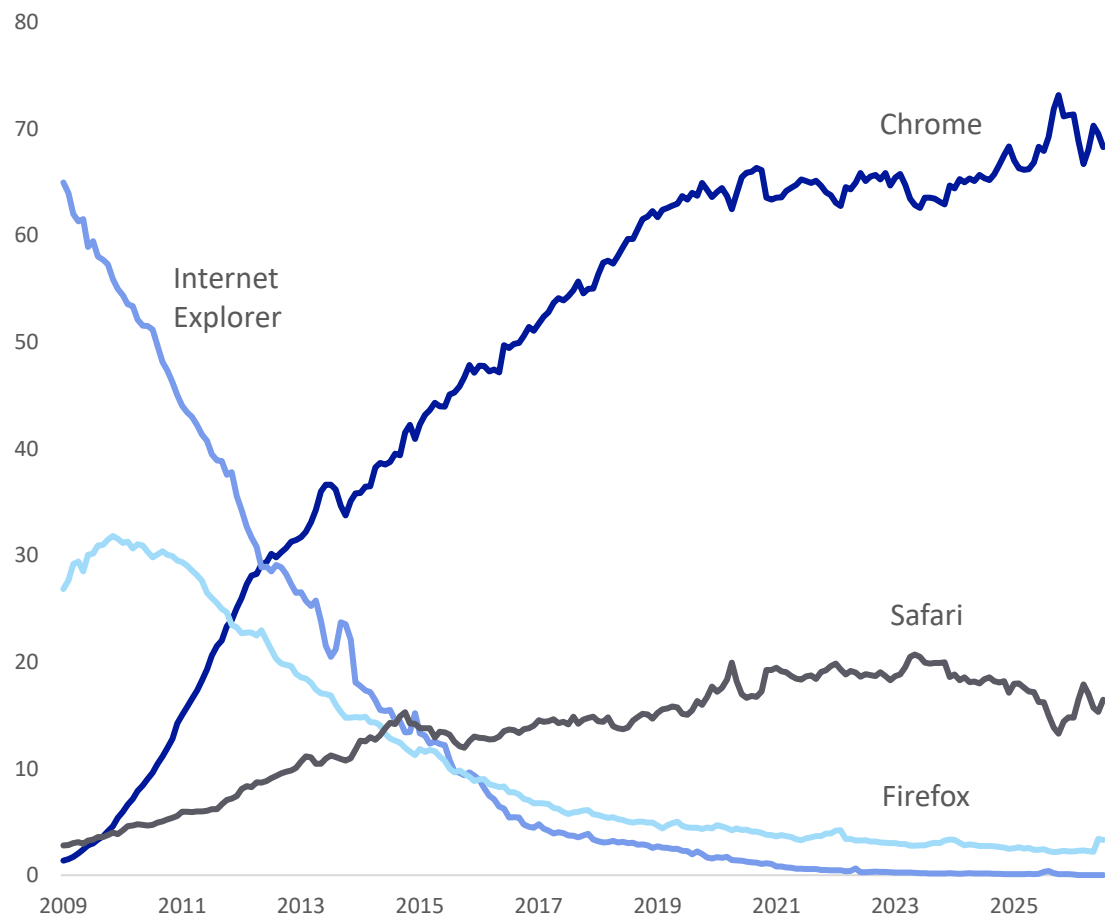
Source: Epoch, Bloomberg Finance LP, The Information (Feb 2026, citing recent updates to investors from OpenAI and Anthropic), Deutsche Bank Research. Note: Year 0 is first year revenue was approx \$10bn (2025 USD)

13. Early leads do not guarantee long-term winners

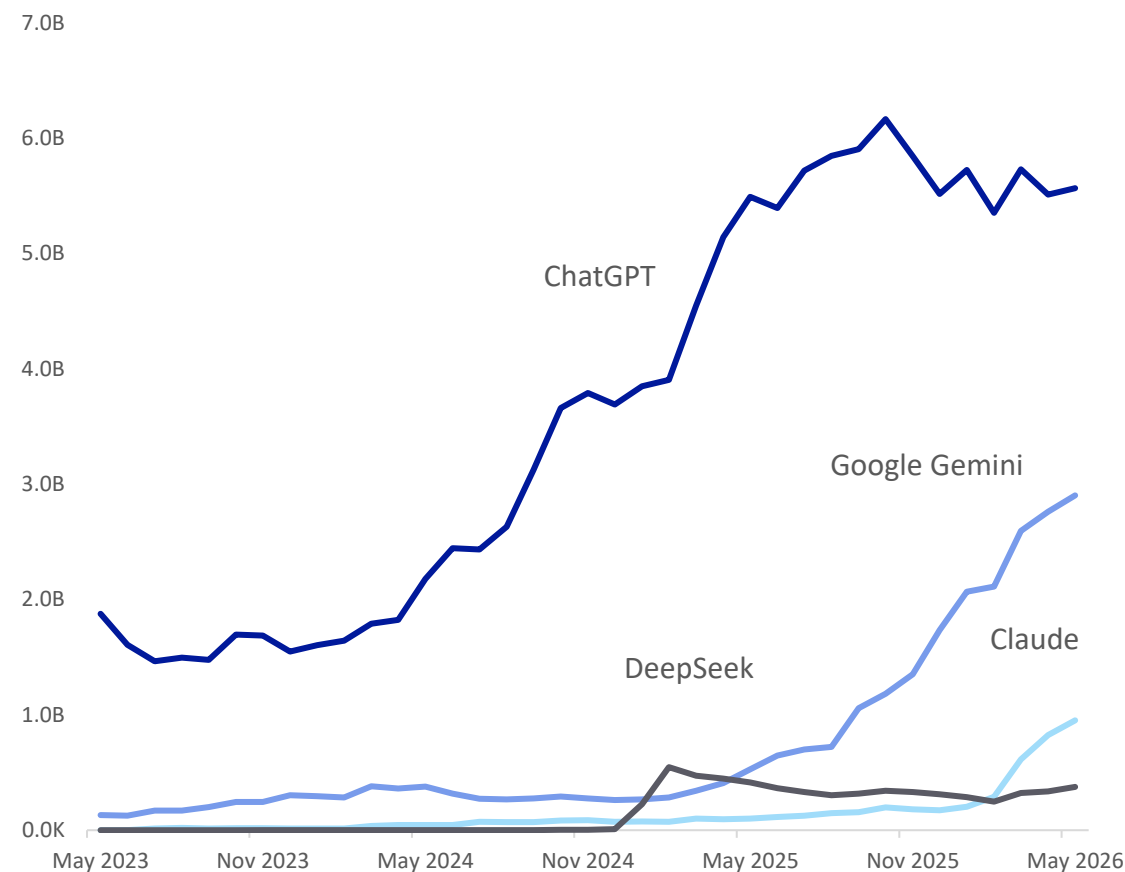
For example, top browser Internet Explorer overtook Netscape, then it too was overtaken



Browser market shares (%) (2009 to today)



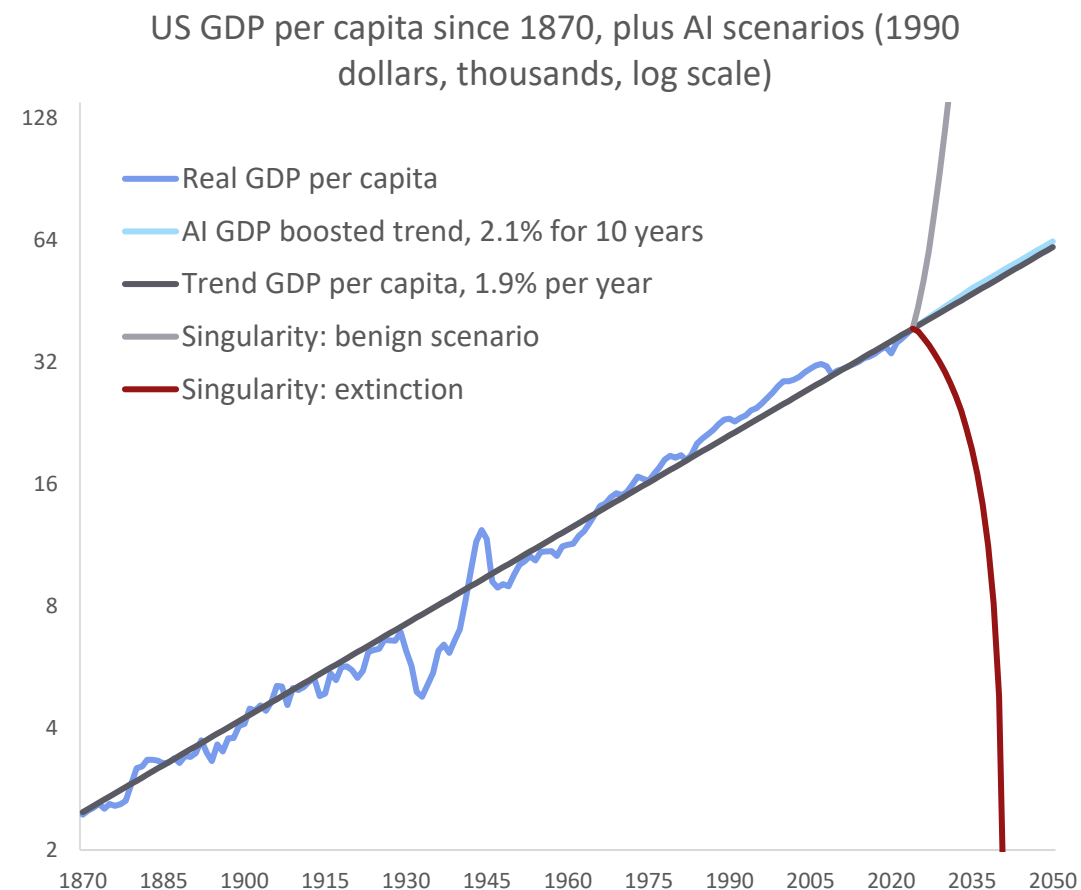
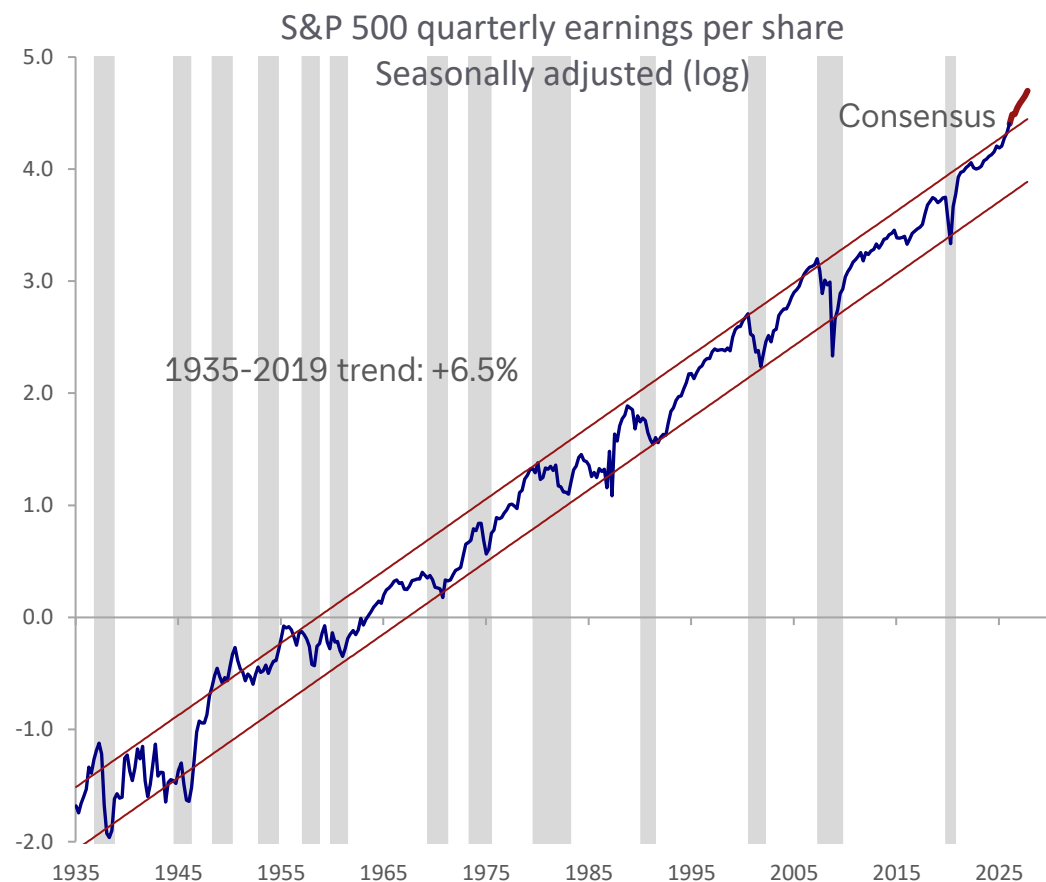
Top four generative AI platforms: monthly visits worldwide (desktop and mobile)



Source: Statcounter, Similarweb, Deutsche Bank Research

14. Long-term trends tend to persist through wars, recessions – and tech revolutions

Even so, some are considering more extreme scenarios this time... from utopia to extinction



Source: Bloomberg Finance LP, Federal Reserve Bank of Dallas, Deutsche Bank Research. Grey shaded bars in S&P quarterly earnings chart represent recessions



Appendix 1

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Deal or no deal, investors are both concerned and excited

August 11, 2026

Executive summary

Five months into the US-Iran conflict and the strength of corporate earnings makes it seem like the optimists have won. But things are more complicated. This report analyses proprietary data from our dbDataInsights team and shows that the issue of investing against the backdrop of a volatile energy market is highly polarising for people. Indeed, of all the investment themes we track, energy has one of the highest rates of both optimism and pessimism.

So, we have an odd situation: On one hand, global corporate earnings are incredibly strong despite the energy shock this year. On the other hand, the Iran conflict is still a massive problem for the world economy. An amicable agreement thus remains elusive.

Three findings from our analysis stand out.

- Energy – whether measured through prices or transition – is now one of the most-cited investment themes for US investors behind AI, and is prompting portfolio changes on par with technology and geopolitics.
- Energy prices carry one of the highest perceived risks of any theme we tested among UK investors.
- Our megatrend model shows that energy disruption (as a medium-term trend) has remained a positive force for the world economy this year. That pattern closely echoes the 1970s oil shocks.

Energy shocks are unambiguously bad for consumers and growth in the short term. But over longer time frames, our model shows that energy disruption has tended to be a net positive for markets and economies because it accelerates diversification of supply and pushes economies to generate more GDP per unit of energy consumed. We see early evidence of a similar dynamic beginning today.

Authors

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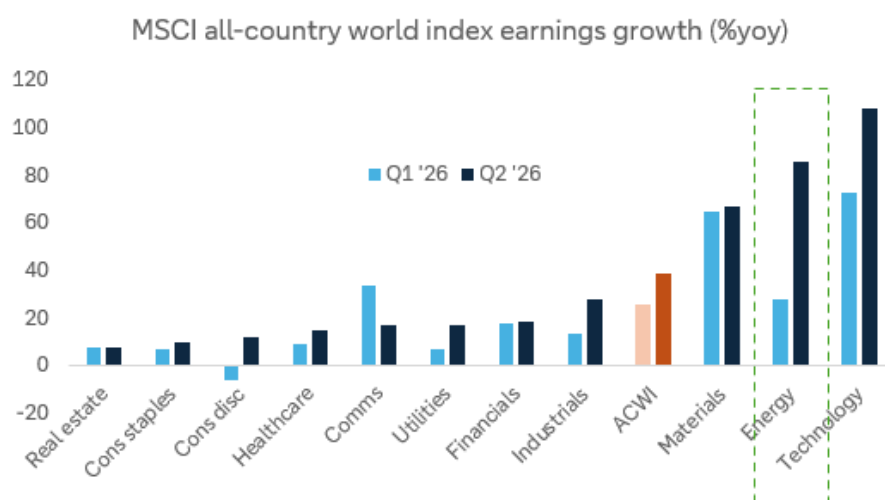
[Galina Pozdnyakova](#)
Research Analyst



A serious issue ... but surprising resilience

The closure of the Strait of Hormuz has not been the economic catastrophe that many expected five months ago. In fact, corporates earnings have just had one of their best quarters outside of recession recoveries. For the companies in the MSCI all-country world index, median earnings growth is running at 16.5% yoy in Q2, well above the 14.7% in Q1. Energy has been a big part of that, with earnings growth, at a sector level, surging 86% in Q2, about triple the growth recorded in the prior quarter.

Figure 1: Earnings growth has been very strong in Q2



Source: Bloomberg Finance L.P., Deutsche Bank Asset Allocation

Economies have fared less well than companies, but they remain remarkably resilient given the harsh predictions that came from some quarters in early March. Of course, stockpile releases and other buffers have helped smooth the disruption – and as these wear down further, larger problems may arise.

So, now we have a curious juxtaposition of outlooks. On one hand, corporate earnings are proving resilient and the long-term focus on energy security has helped make many of the associated equities good investments. On the other hand, the Iran conflict is still a massive problem for the world economy.

What investors think about the effect of energy on their portfolio

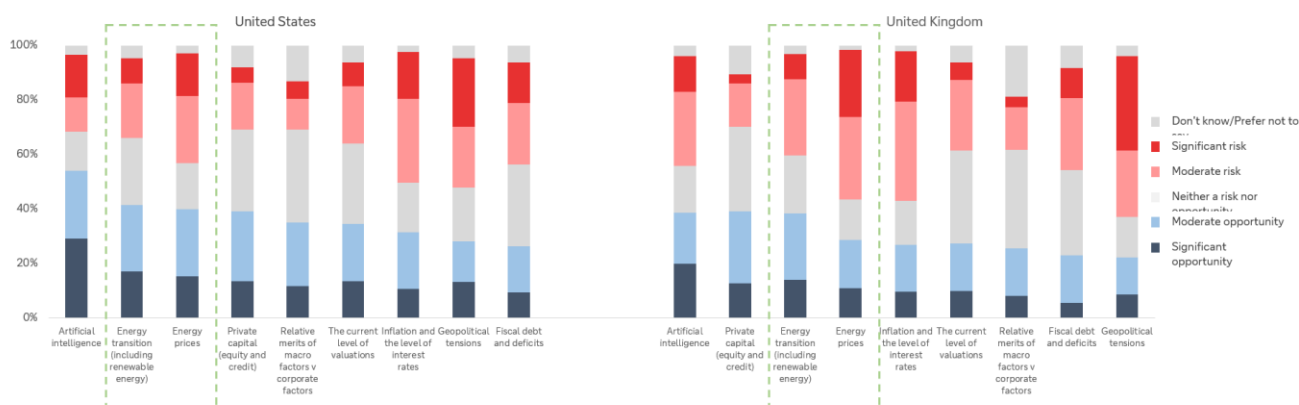
Our dbDataInsights survey on the topic was carried out in April and May of this year and polled a representative sample of people in the US and the UK. The responses indicate that energy has become one of the two dominant investment themes of 2026 – alongside, and closely intertwined with, artificial intelligence.



Energy is a high-conviction theme – in both directions

When we asked investors whether they see various investment themes as a risk or an opportunity for 2026, energy transition and energy prices ranked just behind artificial intelligence as the most-cited opportunities in both the US and the UK. The below chart shows a selection of the themes.

Figure 2: Survey question: "For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



Source: dbDataInsights, Deutsche Bank

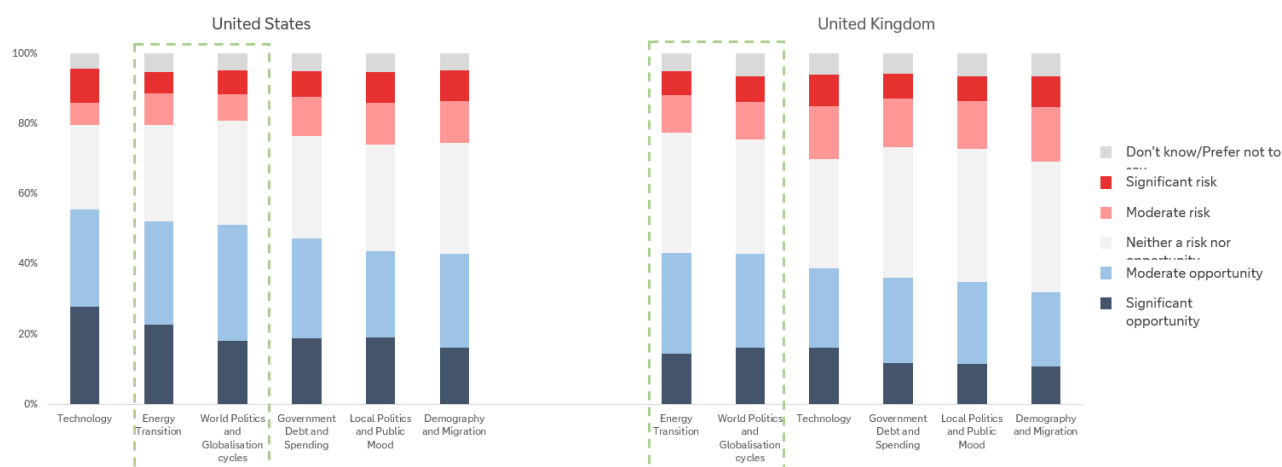
Energy is already changing portfolios

Conviction about energy appears to be translating into action – particularly when we think about it from a medium- or long-term point of view. To assess this, we fell back on our [megatrend model](#) that assesses the development of the six key megatrends that are driving the world's economies and markets.

When we asked investors how likely they are to change their portfolio over the next 12 months because of each megatrend, a dominant theme was energy and geopolitics as the below chart shows.



Figure 3: Survey question: “Within the next 12 months, how likely are you to make changes to your investment portfolio due to the following thematic megatrends?”

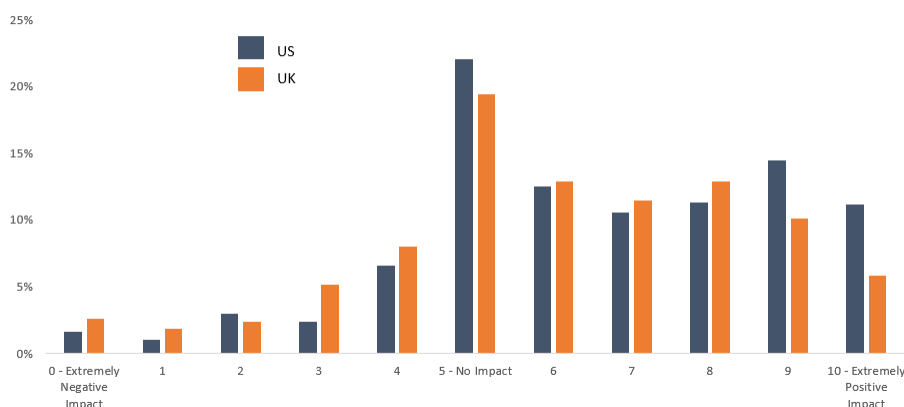


Source: dbDataInsights, Deutsche Bank

Taken together, the two survey questions tell a consistent story: energy has moved from a background, structural megatrend to one that investors are actively trading around, much as technology has been for the AI boom. The difference is that, unlike technology, energy's near-term path is being set as much by missile strikes and shipping insurance premia as by fundamentals.

Investors appear to sense this: when we asked how the energy transition will affect their main portfolio over the next three years, responses skewed firmly positive.

Figure 4: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years



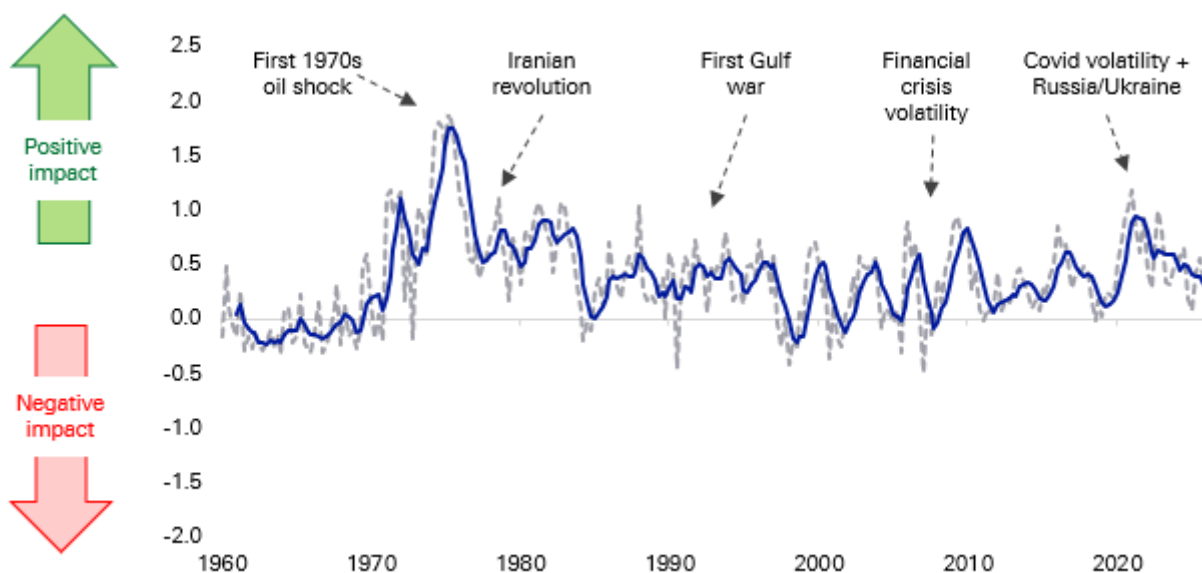
Source: dbDataInsights



Energy as a megatrend: why the medium-term signal is so positive

Energy is one of the [six long-term trends](#) that feeds into our megatrend model that uses almost 100 data series since 1957 to track their impact on economies and markets. Energy is a nuanced megatrend as, unlike several others, there is a big difference between the effect of short-term disruption and long-term gain. Indeed, prior periods of energy disruption have been incredibly good for the long-term energy trend, even if the short-term disruption was incredibly painful for consumers and economies.

Figure 5: Our energy megatrend indicator has generally been positive as the world's economies diversify their supplies at the same time that innovation makes the economy more energy efficient

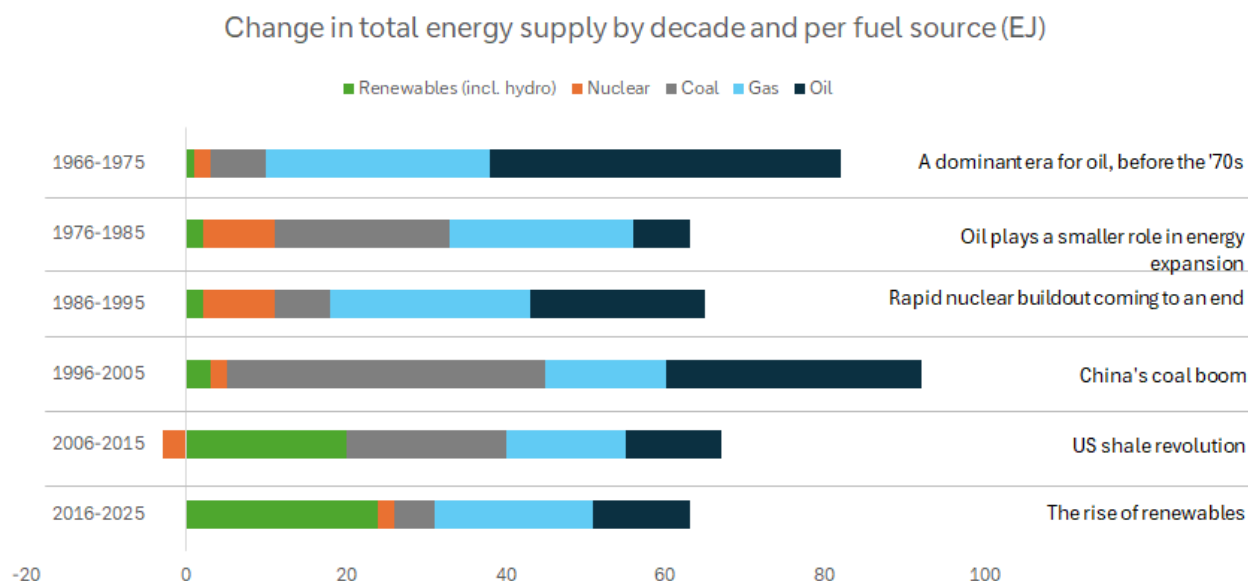


Source: Factset, Bloomberg Finance L.P., Fineon, EIA, Deutsche Bank Research

Viewed through a medium-term lens, the last five months of conflict in the Middle East can be viewed as likely being a long-term positive for economies and markets. That is because the Iran war is accelerating exactly the kind of diversification our model rewards: European and Asian buyers have moved further into US, Guyanese and West African crude and LNG since the Strait's closure; several G20 governments have fast-tracked strategic reserve releases and non-Gulf supply agreements; and renewed urgency has attached to the 'friend-sufficient' push for critical minerals and energy inputs that began during the US-China strategic rivalry.



Figure 6: Shocks over the decades have defined several eras for the evolution of the energy mix



Source: Energy Institute

The 1970s is something of a parallel

The closure of the Strait of Hormuz in 2026 and the disruption to Russian gas flows into Europe in 2022 both echo the twin oil shocks of 1973 and 1979, when OPEC's move to set its own prices followed by the Iranian revolution, sent crude to ten times its early-1970s level and permanently altered the trajectory of global oil demand.

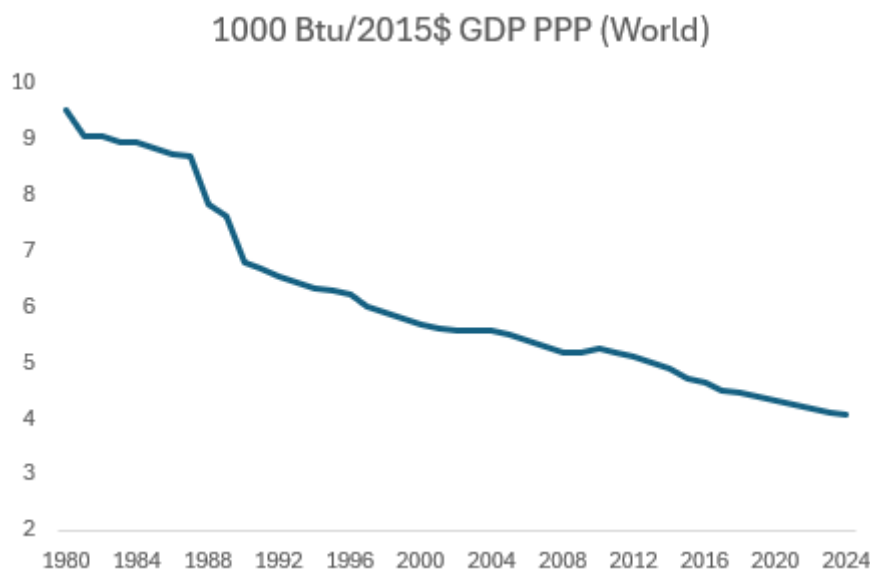
Global oil consumption growth slowed from 7.8% before 1973 to 2.2% pa for most of the rest of the decade (on calculations from the Energy Institute). It outright declined after the second and took a decade for global oil consumption to recover to its 1979 level.

Whilst the 1970s shocks created havoc for both individuals and economies, they accelerated the pace at which the world changed how it generated economic output. The energy intensity of GDP fell 3.3% in 1980 (the largest single-year improvement in the Energy Institute's data series). That adjustment reflected both behavioural change and, over the following decade, structural investment in fuel efficiency and the phase-out of oil-fired power generation.

Since 1980, the energy intensity of GDP has roughly halved as shown in the following chart.



Figure 7: The energy intensity of GDP has been steadily falling – partly because of efficiency and innovation



Source: EIA

This trend is captured in our megatrend model which shows the medium-term payoff from decades of energy disruption. Those forces continue today. Every episode of chokepoint risk, from Ukraine in 2022 to Hormuz in 2026, adds another increment to the diversification of global energy supply and another incentive for energy innovation across economies.

Conclusion

Five months into the US-Iran conflict war, investors are highly polarised about the investment merits of investing against the backdrop of energy volatility. However, those that are positive are very positive. Our megatrend model suggests that reaction is rational: like the 1970s oil shocks, the medium-term legacy of this conflict is likely to be faster energy diversification and a further, durable improvement in the energy intensity of GDP, even though the near-term path for prices is likely to stay volatile.

As a result, energy security and supply diversification should keep attracting capital regardless of where the oil price settles. The beneficiaries are likely to cascade up and down the value chain including, LNG infrastructure, non-Gulf production, renewables, strategic reserves and grid resilience as countries and companies all try to diversify their chokepoint risk.



Appendix 1

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Korea: From Industrialization to Internationalization

August 10, 2026

Executive Summary

Korea sits at the intersection of some of the most important structural forces reshaping the global economy. Re-industrialization, AI infrastructure investment and supply-chain diversification are directing capital toward industries where Korea already possesses scale, technological leadership and established industrial ecosystems. At the same time, policymakers are reinforcing these advantages by supporting strategic industries and expanding Korea's industrial footprint globally.

Yet a paradox remains. Korea's economic influence has expanded much faster than its financial influence. Despite accounting for roughly 2% of global GDP and playing a central role in several strategic industries, the Won remains only a modestly used international currency and Korean assets continue to trade at valuation discounts. Korea's internationalization agenda seeks to narrow this gap, building a financial system that better reflects the country's growing importance in global trade, manufacturing and technology.

In this report, we argue that:

- **The world is increasingly investing in industries where Korea already leads**, as re-industrialization, AI investment and supply-chain diversification redirect capital toward Korea's areas of strength.
- **Korea already possesses what others are trying to build**, with its strengths in manufacturing, technology, innovation and competitiveness.
- **Internationalization is the next stage of development**, helping Korea retain and intermediate a larger share of the wealth it creates.
- **Korea is pursuing a distinct and phased model of internationalization**, combining market-access reforms, greater offshore usability and gradual liberalization while preserving financial stability.
- **The key market implication is a gradual narrowing of the gap between Korea's economic footprint and financial footprint**, as stronger industrial performance increasingly translates into deeper capital markets, a more internationally used currency and stronger demand for Korean assets.

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The world is investing in the industries where Korea already leads

Industrial policy has returned to the centre of global economic strategy. Across the US, Europe, Japan, China and much of Asia, governments are deploying subsidies, tax incentives, localisation requirements and strategic investment programmes to strengthen domestic manufacturing, secure access to critical technologies and reduce supply-chain vulnerabilities. While the policies differ across jurisdictions, the direction of travel is increasingly aligned - resilience, technological leadership and economic security now rank alongside efficiency as core policy objectives.

Figure 1: Industrial policy comparison — At-a-Glance

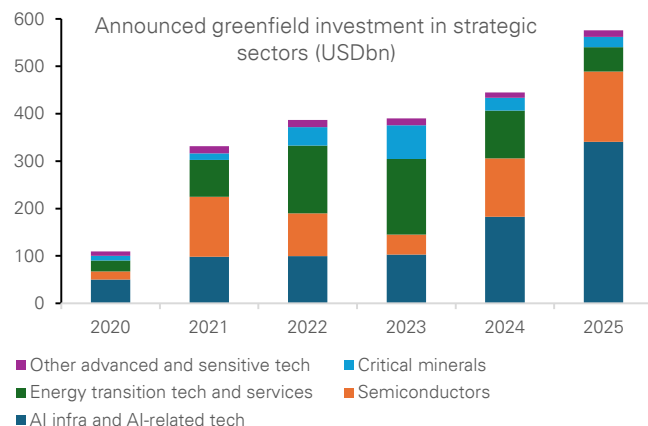
	Policy direction	Strategic sectors	Main objective	Incentive model
Advanced economies				
US	Reshore critical technology production	Chips; EVs/batteries; clean energy; defence; critical minerals	Secure strategic supply chains with trusted partners	Subsidies, tax credits, grants, loans, export controls
EU	Make decarbonisation an industrial growth engine	Net-zero industry; clean tech; heavy industry; raw materials	Build competitive clean industry while protecting competitiveness	State-aid flexibility, EU funding, carbon rules, faster permits
Japan	Strengthen economic security and low-carbon industry	Chips; batteries/EVs; defence; aerospace; nuclear/clean energy	Secure advanced inputs and reinforce defence/energy resilience	Targeted subsidies, tax breaks, procurement, energy investment
North Asia				
China	Pursue self-sufficiency and technology leadership	Chips; EVs/batteries; shipbuilding; green tech; critical minerals	Reduce external vulnerability while preserving export strength	Subsidies, cheap credit, tax relief, procurement, export controls
Korea	Scale national champions in strategic technologies	Memory chips; AI; EV batteries; autos; shipbuilding; defence; nuclear	Hold top-tier chip position and diversify growth engines	Mega capex funds, policy credit, tax credits, R&D and talent programmes
Taiwan	Extend semiconductor leadership into AI hardware	IC design/manufacturing; AI servers; smart manufacturing; digital sectors	Use AI + chips to deepen the innovation ecosystem	R&D credits, shared labs, pilot lines, talent/startup support
South Asia				
India	Scale manufacturing and domestic value-add	PLI sectors; electronics; autos/EVs; chips; pharma; capital goods	Become a high-value manufacturing and export hub	Output-linked incentives, sector schemes, infrastructure/logistics funding
Indonesia	Move nickel downstream into EV value chains	Nickel; battery materials; cells; EV assembly; clean-energy metals	Shift from raw materials to integrated battery/EV production	Export bans, smelter controls, industrial parks, state-firm ecosystems
Malaysia	Upgrade from assembly to higher-value manufacturing	E&E; semiconductor back-end; advanced materials; aerospace; pharma	Move up technology and Industry 4.0 capability	Tax incentives, grants, roadmaps, cluster facilitation
Philippines	Revive manufacturing around electronics and components	Electronics packaging; autos; aerospace components; broader manufacturing	Deepen regional supply-chain participation beyond services	SEZ incentives, FDI promotion, sector roadmaps, streamlined approvals
Singapore	Anchor high-value advanced manufacturing	Precision engineering; chips; pharma; robotics; automation; digital manufacturing	Keep high-value production linked to R&D and HQ functions	Grants, co-funding, R&D incentives, skills schemes, bespoke EDB packages
Thailand	Use EEC and S-curves to upgrade industry	EVs; smart electronics; aviation/logistics; robotics; bio/medical industries	Become a regional hub for upgraded manufacturing	SEZ tax breaks, infrastructure, premium sector incentives
Vietnam	Build supporting industries and semiconductor capability	Electronics support; mechanical/autos; textiles/footwear; high-tech; chips	Raise localisation and integrate deeper into global supply chains	Cost-sharing subsidies for R&D, technology and workforce spending

Source : Deutsche Bank Research

The scale of this shift is already visible in global investment flows. According to [UNCTAD](#), strategic sectors—including semiconductors, clean energy, advanced manufacturing, defence and digital infrastructure—accounted for 44% of global greenfield investment in 2025, up from just 16% in 2020. Over the same period, the value of announced projects rose from \$109bn to \$576bn, highlighting the growing concentration of capital in strategic industries. This is creating one of the largest industrial investment cycles in decades and directing investment towards precisely the sectors where Korea has already established globally competitive capabilities.

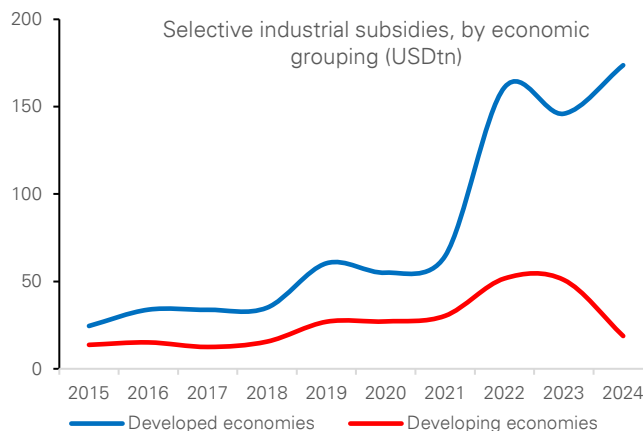


Figure 2: Growing investments in strategic sector globally



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD). Note: Data for 2025 are annualized on the basis of information available as of 30 November.

Figure 3: Increase in the use of subsidies to support growth in the strategic sector



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD), based on New Industrial Policy Observatory (NIPO). Note: Only subsidies worth more than \$10mn are recorded in the NIPO database.

Three structural forces are now converging in Korea's favour.

- **Reshoring, ally-shoring and rearmament are increasing global demand for industries in which Korea already holds a competitive advantage.** Semiconductors, shipbuilding, defence equipment, batteries and critical minerals have become central to national-security and supply-chain strategies. Post-Ukraine European rearmament has increased demand for reliable defence suppliers, while the US and its allies are seeking trusted partners to rebuild strategic supply chains. Korea already possesses many of the industrial capabilities that advanced economies are now attempting to reshore or source from aligned partners.
- **The global AI capex supercycle is providing a major terms-of-trade uplift.** Hyperscaler capital expenditure is estimated to be growing at close to 70% YoY through 2024-26, creating strong demand for GPUs, HBM, DRAM and related semiconductor infrastructure. As large language models become increasingly complex and agentic AI requires greater processing power and speed, demand for advanced memory chips is rising sharply. Samsung Electronics and SK Hynix place Korea at the centre of this cycle, positioning the country as one of the most direct beneficiaries of the global AI infrastructure buildout.
- **China de-risking is reshaping supply chains in Korea's favour.** Western buyers are actively reducing reliance on Chinese supply chains across selected strategic sectors, including batteries, battery materials, vessels, robotics and critical minerals. Korea is one of the few economies with both the industrial scale and geopolitical alignment to absorb some of the production, investment and market share being redirected away from China. While Chinese overcapacity remains a competitive challenge in some industries, the global effort to diversify away from China is simultaneously creating opportunities in strategic sectors where technology, reliability and security considerations matter more than cost alone.



This convergence matters. Korea is not benefiting from one isolated export cycle. **Rather, it sits at the intersection of ally-shoring, AI infrastructure investment and China supply-chain diversification—three structural forces that are likely to persist for years.** In many respects, the world is increasingly investing in precisely the industries where Korea already possesses scale, technological leadership and established industrial ecosystems.

Figure 4: Korea ranks among the world's best positioned economies for re-industrialisation

Rankings (1 = Best)													
Metric	US	EU	Japan	China	Korea	Taiwan	Singapore	Malaysia	Indonesia	Thailand	Vietnam	Philippines	India
Manufacturing value added (% of GDP)	13	11	8	3	2	1	9	6	7	5	4	10	12
Share of high-tech exports	11	10	7	8	2	1	3	6	13	9	5	4	12
Current account balance (% of GDP)	13	8	5	6	4	1	2	9	10	7	3	12	11
Government debt (% of GDP)	12	10	13	11	5	2	8	4	7	3	6	9	12
Gross capital formation (% of GDP)	11	13	6	1	5	7	10	9	3	12	4	8	2
Total surplus capacity (MW)	3	2	5	1	6	10	13	12	7	9	8	11	4
Electricity price (USD/kWh)	8	12	11	4	5	10	13	7	1	6	2	9	3
Business infrastructure	1	7	6	3	5	4	2	8	12	9	10	13	11
Global talent competitiveness index	3	4	5	8	6	2	1	7	12	11	10	9	13
Regulatory quality	2	5	3	11	6	4	1	7	8	9	13	10	12
Trade freedom	12	6	8	9	10	2	1	3	7	11	5	3	13
Broad competitiveness	3	10	9	4	6	2	1	5	13	7	8	12	11
Economic Complexity	6	7	1	5	3	2	4	8	13	9	12	11	10

Source : Deutsche Bank Research, CEIC, Haver Analytics, Insead, World Bank, The Heritage Foundation, IMD.org, Harvard Business School

Policy is also reinforcing these advantages. Rather than creating entirely new industries, Korean policymakers are largely focused on scaling existing strengths, securing critical inputs and supporting the global expansion of Korean industrial ecosystems. Recent initiatives have directed capital toward semiconductors, AI infrastructure, batteries, defence and other strategic sectors through policy financing, investment programs and regulatory support. At the same time, Korea is strengthening access to critical minerals, expanding overseas industrial partnerships and facilitating outward investment by Korean firms. The objective is not simply to support national champions, but to reinforce the broader ecosystem around them and ensure Korea remains embedded in the next phase of global industrial development.

Korea already possesses what others are trying to build

Few economies are better positioned to benefit from the new industrial order than Korea. Unlike many countries attempting to rebuild manufacturing capacity from a relatively low base, Korea already has one of the world's most advanced industrial ecosystems.

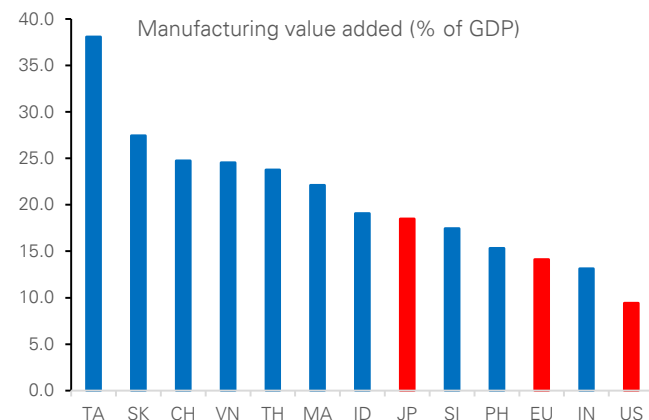
Few economies are better positioned to benefit from the new industrial order than Korea. Unlike many countries attempting to rebuild manufacturing capacity from a relatively low base, Korea already has one of the world's most advanced industrial ecosystems.

- **Industrial ecosystems matter.** Korea has built globally competitive positions across semiconductors, batteries, shipbuilding, defence, robotics, nuclear energy and advanced manufacturing. These industries are supported by a deep industrial ecosystem spanning engineering talent, R&D capabilities, specialised supplier networks and world-class infrastructure and relatively healthy public and external balances. These capabilities are difficult to replicate quickly and increasingly valuable in a world prioritising resilient and secure supply chains.



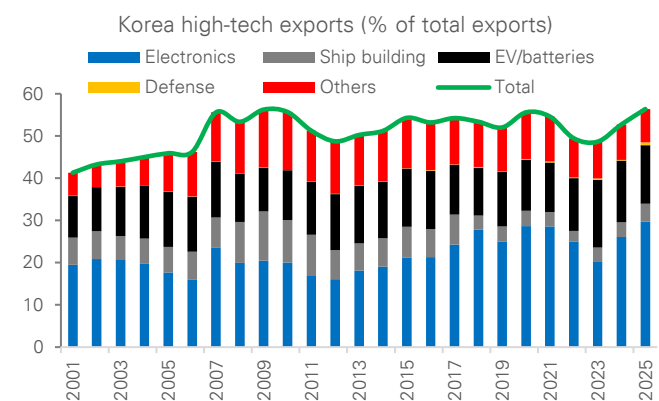
- **Starting from strength matters.** Manufacturing accounts for approximately 24.3% of GDP, among the highest shares in the OECD and roughly two times the level of the advanced economies. At a time when governments are seeking to expand domestic production of strategic goods, Korea enters the re-industrialisation cycle with an existing manufacturing base rather than having to build one from scratch.
- **Technology leadership matters.** Approximately 58% of Korea's exports are high-tech products, one of the highest shares among major economies, while semiconductors account for roughly 50% of high-tech exports. Korea also accounts for ~60% of global memory semiconductor production and 90% of global HBM supply, making it one of the most important beneficiaries of the AI infrastructure cycle.

Figure 5: Manufacturing accounts for approximately 24.3% of GDP, among one of the highest shares in the OECD and roughly two times the level of the advanced economies.



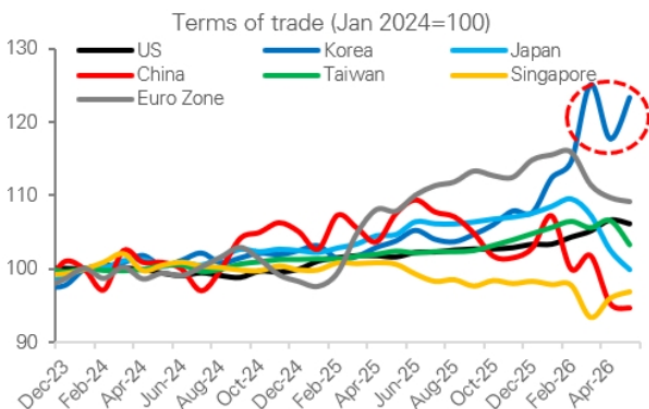
Source : Deutsche Bank Research, Haver Analytics

Figure 6: Korea's export engine increasingly driven by High-Tech industries



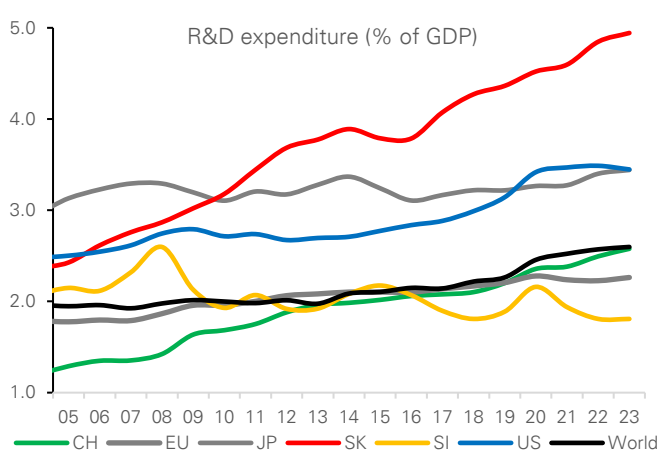
Source : Deutsche Bank Research, ITC

Figure 7: Significant improvement in terms of trade since 2024 for Korea



Source : Deutsche Bank Research, Bloomberg Finance LP, Haver Analytics

Figure 8: Korea has one of the highest R&D expenditures in the world



Source : Deutsche Bank Research, World Bank



- **Strategic industries matter.** Beyond semiconductors, Korea has built globally competitive positions in EV batteries, defence, advanced shipbuilding, robotics and nuclear energy. Korean firms account for ~30% of global [EV battery capacity](#) and 90% of global [LNG carrier orders](#), while defence exports are increasingly benefiting from rising global security spending and rearmament efforts.
- **Energy increasingly matters.** Under South Korea's [latest electricity plan](#), nuclear power is expected to account for approximately 35% of electricity generation by 2038, while carbon-free sources are projected to supply roughly 70% of total generation. The plan is designed in part to accommodate rising electricity demand from semiconductors, AI data centres and broader industrial electrification. As manufacturing and digital infrastructure become increasingly energy-intensive, access to reliable, scalable and lower-carbon power is emerging as an important component of industrial competitiveness.

Korea's advantages extend beyond individual industries. Through leadership in semiconductors, batteries, shipbuilding, defence and advanced manufacturing, Korea is becoming an increasingly important industrial partner for both advanced and emerging economies. As supply chains become more closely linked to economic security, reinforcing its role within the evolving architecture of the global economy. Rather than acting solely as an exporter of goods, Korea is increasingly participating in the development and expansion of industrial ecosystems across multiple regions through investment, technology transfer and supply-chain partnerships. In that sense, **Korea's industrial strengths are not only supporting growth and competitiveness at home but are also reinforcing its role within the evolving architecture of the global economy.**

Korea's industrial position remains strong, but its advantages should not be taken for granted. Despite leadership in semiconductors, shipbuilding, defence and advanced manufacturing, Korea remains dependent on imported energy, critical minerals and raw materials, leaving parts of its industrial ecosystem exposed to commodity and supply-chain shocks. Competition is also intensifying. China continues to move up the value chain across batteries, shipbuilding, robotics and segments of the semiconductor supply chain, while governments globally are deploying increasingly aggressive industrial policies to build domestic capabilities. Industrial leadership is rarely permanent. **Korea's re-industrialisation story therefore depends not only on existing strengths, but also on its ability to sustain innovation, secure critical inputs and preserve its technological edge over the coming decade.**

Internationalization as the next stage of development

Korea scores strongly on measures of industrial depth, export sophistication and external strength. Manufacturing value added, high-tech exports and current-account balances compare favourably with most regional peers. Yet Korea remains among the cheapest major Asian markets on P/E and P/B metrics, while the Won screens as undervalued on our valuation framework.



Figure 9: Korea's valuation paradox: strong fundamentals, cheap assets

	Economic Matrix			Financial Market Matrix				
	Manufacturing value added (% of GDP)	Share of high-tech exports (%)	Current account bal (% of GDP)	12M forward P/E ratio	ROE (%)	12M forward P/B ratio	Dividend yield	FX valuation
US	9	28	-4	21.3	20.0	4.6	1.2	5.7
EU	14	31	2	15.8	12.6	2.2	3.1	-1.6
Japan	18	40	5	22.4	12.0	2.7	1.5	-24.7
China	25	35	4	14.9	9.9	1.6	2.6	-5.3
Korea	27	59	7	6.9	10.5	1.5	1.9	-15.0
Taiwan	38	78	17	20.3	13.9	4.0	2.1	-5.3
Singapore	17	55	17	17.2	9.9	1.8	4.1	4.9
Malaysia	22	48	2	15.1	10.6	1.5	4.3	1.8
Indonesia	19	9	0	10.0	11.8	1.3	6.3	-7.8
Thailand	24	31	3	16.1	8.6	1.6	3.6	7.5
Vietnam	25	48	7	12.0	15.1	1.7	1.9	
Philippines	15	53	-3	9.9	11.6	1.2	3.7	2.9
India	13	23	-1	19.9	14.2	2.6	1.8	-6.6

Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

Source: Deutsche Bank Research, CEIC, Haver Analytics, Bloomberg Finance LP. Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

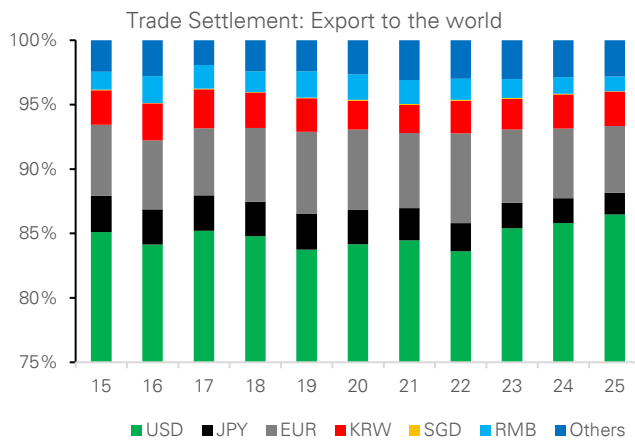
Our preferred valuation framework suggests the Won remains materially undervalued relative to fundamentals. Korea's current account surplus could reach ~10.3% of GDP this year and remain above 5% over the medium term, well above its long-run average of around 3.5% of GDP. Under a FEER framework, which anchors exchange rates to external-balance sustainability, such an external position is inconsistent with current KRW levels and **implies that the Won is undervalued by ~ 10% in trade-weighted terms.**

More importantly, this result may prove conservative. Korea's current account improvement increasingly reflects a positive terms-of-trade shock driven by AI-related semiconductor demand, rising export sophistication and stronger pricing power. As a result, Korea's industrial renaissance may not only justify larger external surpluses, but also a higher equilibrium valuation for the Won.

Besides asset valuation discounts, Korea's growing economic importance is not reflected in its financial system and currency yet. Today, Korea accounts for roughly 2% of global GDP, ranks among the world's largest exporters of advanced manufactured goods and is home to several globally strategic industries. Won accounts for approximately 0.9% of global FX turnover, which puts it at around 12th rank; while its role in international payments, funding markets and reserve holdings remains considerably smaller than Korea's relative economic quantum.

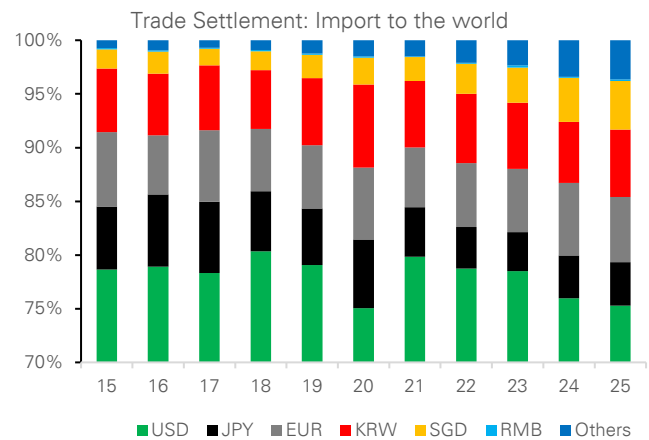


Figure 10: Korea's exports are still overwhelmingly settled in foreign currencies



Source : Deutsche Bank Research, Haver Analytics

Figure 11: The won plays only a limited role in Korea's import settlement



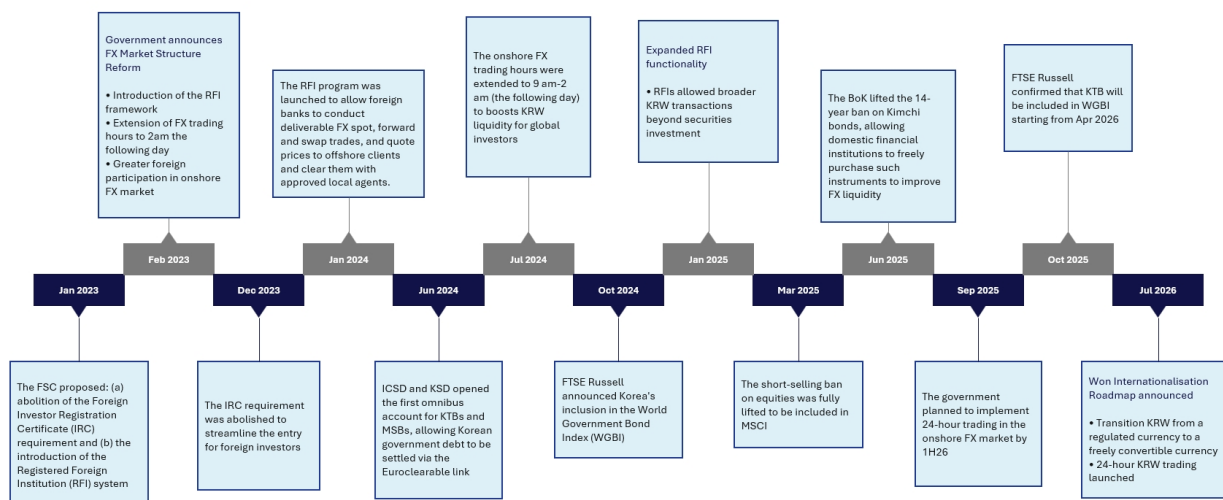
Source : Deutsche Bank Research, Haver Analytics

This gap between economic importance and financial influence helps explain the government's newly [announced](#) roadmap for Won internationalization. The objective is not to transform the Won into an alternative reserve currency, nor to replicate China's offshore RMB model. Rather, policymakers are focused on creating the conditions for a more internationally usable currency — one that is easier to trade, fund, settle and hold across time zones. The roadmap combines the development of offshore settlement and funding infrastructure with a gradual easing of selected capital-account restrictions, allowing international use of the Won to expand while maintaining a measured and risk-managed approach to liberalization.

More fundamentally, Korea's push to internationalize the Won reflects a broader question about how the country captures the benefits of its industrial success. Our work argued that re-industrialization, the AI infrastructure cycle and supply-chain diversification are likely to strengthen Korea's external balances and generate larger pools of national savings. Yet generating wealth and retaining wealth are not the same thing. A significant share of the savings generated by Korea's industrial success continues to be recycled abroad through portfolio investment and overseas asset accumulation. **Strong external balances therefore do not automatically translate into deeper domestic capital markets, stronger asset valuations or a more internationally important currency.**



Figure 12: The reforms started well before July 2026



Source : Deutsche Bank Research

In many respects, Won internationalization can be viewed as the financial counterpart to Korea's industrialization. Industrialization generates exports, profits, savings and external surpluses; internationalization helps deepen capital markets, broaden participation and increase the likelihood that a greater share of the resulting trade, investment and funding activity is intermediated through Korean financial markets. **The objective is therefore not simply a more international currency, but a financial system that more closely reflects Korea's growing role in the global economy.** The result is not a wholesale opening of the capital account, but a gradual and carefully sequenced effort that combines greater offshore usability of the Won with the selective liberalization of domestic financial markets, while preserving financial stability and policy flexibility.

Korea's model – A phased approach to internationalization

A common mistake is to analyze Korea's internationalization strategy primarily through the lens of either China or Singapore. In reality, Korea is pursuing a distinct path that differs from both models.

China's approach relied heavily on the creation of a parallel offshore ecosystem that allowed the renminbi to internationalize while maintaining extensive control over domestic financial markets. Singapore represents the opposite approach, combining deep global integration with virtually unrestricted capital mobility. Korea's strategy sits somewhere between these two models. Policymakers are seeking to expand the usability of the Won while preserving financial stability and policy flexibility.

The first phase of reform focused on improving market accessibility. Measures such as the Registered Foreign Institution framework, omnibus accounts, extended FX trading hours and broader market-access reforms were designed to reduce operational barriers facing global investors and improve access to Korean financial assets. These reforms laid the foundations for greater foreign participation in Korean bond and equity markets.



The next phase goes further by focusing on the usability of the currency itself.

Recent reforms are designed to make the Won easier to hold, transfer, fund and settle internationally. Rather than concentrating solely on attracting investors into Korean assets, policymakers are now seeking to expand the range of activities that can be conducted in Won across investment, funding and settlement activities.

Importantly, liberalization is being implemented gradually. Korea is not pursuing a wholesale opening of the capital account. Instead, reforms are being sequenced carefully alongside monitoring systems, liquidity facilities and macroprudential safeguards. The objective is to reduce frictions and improve accessibility without compromising financial stability.

The result is best understood as a phased transition from market-access reform towards broader currency internationalization. Earlier reforms focused on making Korean financial assets easier for global investors to access. The next phase is aimed at making the Won itself easier to use across investment, funding and settlement activities.

Figure 13: Korea is pursuing a distinct model of currency internationalization

Korea is pursuing a distinct model of currency internationalisation				
	Singapore	China	India	Korea
Primary objective	Global financial centre	Global reserve currency	Expand INR use in trade and finance	Increase KRW investability and global usage
Internationalisation model	Full openness	Offshore-onshore framework (CNH/CNY)	Trade-led internationalisation	Phased internationalisation led by capital-market integration
Key policy tools	Open capital account, global financial intermediation	Offshore RMB markets, clearing banks, Connect programmes	INR trade settlement, bilateral arrangements	FX market liberalisation, benchmark inclusion, settlement infrastructure
Treatment of offshore markets	Integrated	Separate offshore ecosystem	Limited	Greater offshore usability through infrastructure linked to the domestic market
End-state vision	Fully open financial centre	Widely used international currency	Broader regional and trade usage	Globally investable and increasingly usable currency
Key challenge	Small domestic market	Capital controls	Limited global demand	Balancing openness and financial stability
Sequencing	Openness first	Trade settlement → Offshore ecosystem → Investment integration	Trade settlement → Gradual financial liberalisation	Market access → Broader currency usage

Source : Deutsche Bank Research

How Korea is building a more international Won?

The first pillar is market accessibility. Korea has already moved to 24-hour dollar-Won spot trading, allowing investors to trade and hedge Won exposure across Asian, European and North American trading sessions. Earlier reforms such as the Registered Foreign Institution (RFI) framework, omnibus accounts and settlement-system upgrades were designed to reduce many of the operational frictions that historically limited foreign participation in Korean financial markets. These measures laid the foundations for greater foreign ownership of Korean bonds, equities and other financial assets.

The second pillar is offshore usability. The proposed framework allows registered Offshore Won Settlement Institutions to facilitate Won transfers, funding and settlement between non-residents without requiring every transaction to pass through traditional domestic account structures. A new BoK offshore settlement network is intended to support 24-hour settlement and improve the ability of



investors, corporates and financial institutions to hold, transfer and deploy Won outside Korean business hours. Authorities are also seeking to encourage greater use of deliverable KRW markets through offshore settlement infrastructure and measures that support deliverable rather than non-deliverable instruments over time. This reflects one of the central features of the roadmap: expanding the ability of non-residents to hold, transfer, fund and settle Won offshore, even as onshore liberalisation remains gradual and carefully sequenced.

The third pillar is gradual liberalization. Rather than removing all restrictions at once, policymakers are selectively easing reporting requirements, simplifying account structures and gradually shifting certain transactions from prior approval to ex-post reporting. Importantly, liberalization is occurring within a controlled framework: offshore Won transactions initially operate through registered institutions, while market monitoring, liquidity facilities and prudential safeguards are expanded in parallel.

Figure 14: The building blocks of KRW internationalization

Pillar	What has changed	Why it matters
Market accessibility	24-hour USD/KRW trading; RFI framework; omnibus accounts; settlement upgrades	Allows foreign investors to access, trade and hedge Korean assets more easily across global time zones
Capital market integration	WGBI inclusion, MSCI market-access reforms, Euroclear/ICSD connectivity, expanded collateral and repo infrastructure	Reduces operational barriers for global investors and integrates Korean assets more closely with global portfolios
Offshore won usability	Offshore Won Settlement Institutions and offshore settlement network	Allows non-residents to hold, transfer, fund and settle won outside Korea more efficiently
Regulatory liberalisation	Many inter-foreigner won capital transactions move toward lighter reporting and ex-post supervision	Reduces frictions for investors, corporates and treasury centres while maintaining oversight
Digital money infrastructure	Stablecoin framework, CBDC initiatives, tokenised government-bond pilots	Establishes future rails for cross-border payments and digital won usage

Source : Deutsche Bank Research

Taken together, these reforms represent an evolution from market-access reform towards broader currency internationalization. Earlier initiatives such as the RFI framework, omnibus accounts, extended trading hours and benchmark-related reforms focused primarily on reducing barriers for foreign investors. The new roadmap builds on that foundation by expanding offshore settlement infrastructure, easing selected capital-account restrictions and increasing the range of activities that can be conducted in Won. The objective is therefore not only to make Korean assets easier to access, but also to make the Won itself easier to hold, transfer, fund and settle internationally.

Market Implication: A new funding, investment and settlement currency for Asia?

For the currency, success should not be measured by whether the Won becomes a major reserve currency. **A more realistic benchmark is whether the Won gradually plays a larger role in economic activities linked to Korea's own industrial and trading footprint.** Progress would likely be visible through greater use of Won settlement in trade, broader participation by foreign investors in Korean financial markets, and increasing use of Won-denominated funding and hedging instruments. The reforms create the infrastructure necessary for such a transition,



but market adoption will ultimately determine the outcome.

Figure 15: Measuring progress towards a more international Won

Success Metric	Korea	Japan	Australia	Why it matters
Foreign ownership of local-currency government bonds	28.7%	13.7%	52.6%	Higher foreign participation and broader investor base
Foreign ownership of domestic equities	38.5%	11.9%	30.9%	Increased representation in global portfolios
Institutional ownership of equities	20.0%	34.0%	60.0%	More stable portfolio flows
Inclusion in Global DM benchmark	Not in MSCI DM	Yes	Yes	Measures potential structural demand from its financial assets
Home-currency settlement share in external trade	3.0%	33% (exports only)	8.8%	Gradual increase in trade invoicing and settlement in won
Foreign participation in repo / funding markets	2.1%	N/A	60.0%	Deeper offshore funding and collateral markets
Currency share of global FX turnover	0.9%	8.5%	3.1%	Greater turnover and broader participation

Source : Deutsche Bank Research

The most immediate market impact may be a gradual migration from NDFs toward deliverable KRW markets. Improved access and offshore settlement infrastructure should support greater use of deliverable KRW instruments over time, although NDFs are likely to remain an important part of the ecosystem. The goal is not to eliminate NDFs, but to create more attractive deliverable alternatives, deepen liquidity and strengthen the link between offshore activity and Korean financial markets.

The implications may extend beyond the currency itself. A more internationally accessible financial system could support deeper capital markets, stronger liquidity and a broader investor base for Korean assets. Over time, this may help narrow valuation discounts that persist despite Korea's strong industrial position, improving the linkage between the country's economic performance and financial-market performance. In this sense, internationalization should be viewed not merely as an FX-market reform, but as part of a broader effort to align Korea's financial influence more closely with its growing economic and industrial importance.

Ultimately, Korea's post-war success was built on industrialization. Re-industrialization, AI investment and supply-chain diversification suggest that story is far from over. The next decade may be defined by a different question: whether Korea can convert industrial power into financial power. The success of internationalization will therefore be judged not by reserve-currency status, but by whether Korea narrows the gap between its economic footprint and its financial footprint. If industrialization was the first stage of Korea's development story, internationalization may prove to be the second.



Appendix 1

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The Expanding Frontier of Agentic AI: A User's View

August 6, 2026

The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the [Agent Index from LLM Stats](#) for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive

Authors

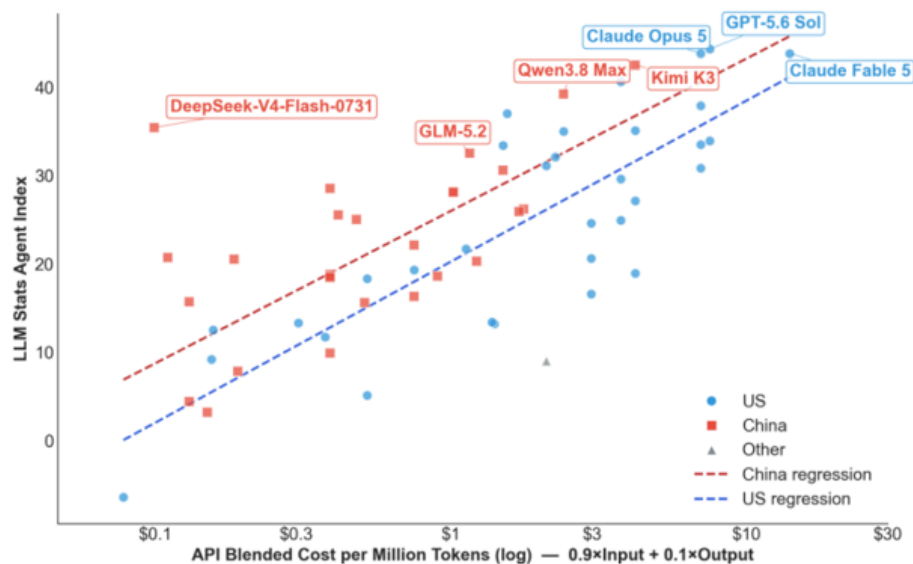
Yi Xiong, Ph.D.
Chief Economist

Deutsche Bank
Research Institute



relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

A frontier moving rapidly

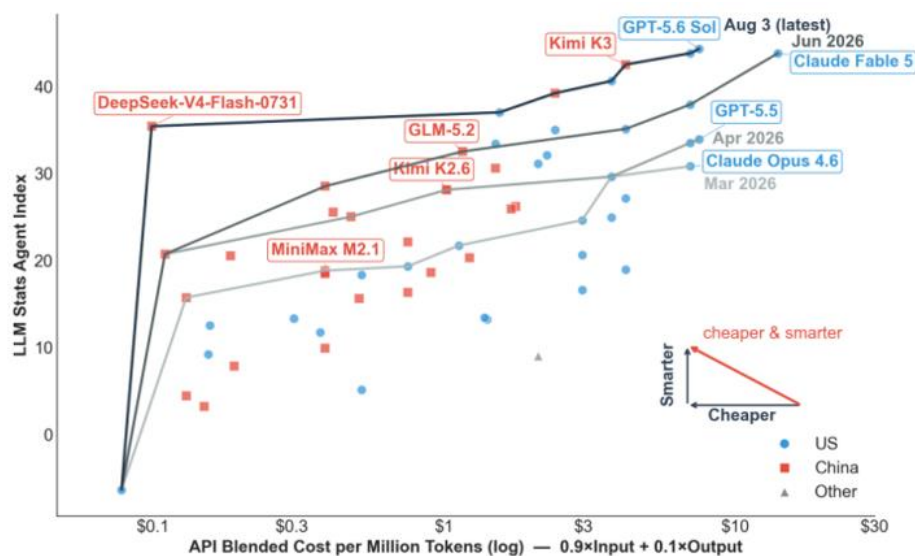
The more useful concept for model selection—at least for me as an economist—is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of \$7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.



Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of \$1.16 is far below Opus 4.7's \$7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead... At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

...but Chinese models' advancement from the lower-left is also remarkable.

In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at \$0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.



DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at \$20/month. A few months later, my recurring spending on AI is now reaching \$100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an \$800 Mac Mini to host my “little lobsters” (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons’ paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me more confident that the future of agentic AI is arriving faster than many expected.**



Appendix 1

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Situational awareness - do we have it?

August 3, 2026

Executive summary

There are two events in the last few weeks that look like non-events. The Situational Awareness hedge fund unwound a \$16bn book of leveraged AI trades with barely a ripple beyond a few days of volatility. Meanwhile, the Kospi fell more than 20% in 48 hours and then rallied 25% off the low — finishing the week almost flat. Taken individually, both look like markets doing their job: absorbing shocks and moving on.

We think that read is wrong, or at least incomplete. Both events were symptoms of the same underlying condition: leverage in ETFs, margin accounts and hedge fund books that was encouraged by more than a decade of low and stable rates. Indeed, reports that over 3% of Korea's adult population received a margin call over the last two weeks is deeply concerning if true.

This piece argues that 'small vol' events are going to affect markets with an increased frequency. That comes as the backdrop of normalised interest rates and geopolitical unrest has been augmented by a Fed that has moved away from the forward guidance that reduced some volatility in the past.

We also worry that the risks in household debt (and investment accounts on margin) is being ignored as the focus stays on the risks in growing sovereign debt. If this is right, 'small vol' episodes are likely to become more frequent and potentially have larger consequences.

In the end, we worry that investors, corporates and policymakers are focused on avoiding the next big financial crisis and, in doing so, may be underestimating the risk in small events that can cascade.

A tale of two 'small vol' events

Markets have spent the last few years braced for another 2008-style event. That vigilance has a blind spot. It trains attention on the big, obvious risks and away from the smaller, everyday ones that can cascade in the right circumstances.

The implosion of the Situational Awareness hedge fund is the first case in point. The fund sold its \$16bn public equity book after highly leveraged trades on AI companies went wrong. Markets absorbed the unwind with only a brief bout of volatility — no contagion, few headlines beyond the financial press.

The second was in Korea. At one point last week, the Kospi fell over 20% in barely 48 hours, before staging a 25% rebound from the low. The headline index ended the week close to flat and broader markets have shrugged it off. But a flat index conceals an extraordinary amount of forced selling and buying beneath the surface, much driven by margin calls on leveraged retail positions.

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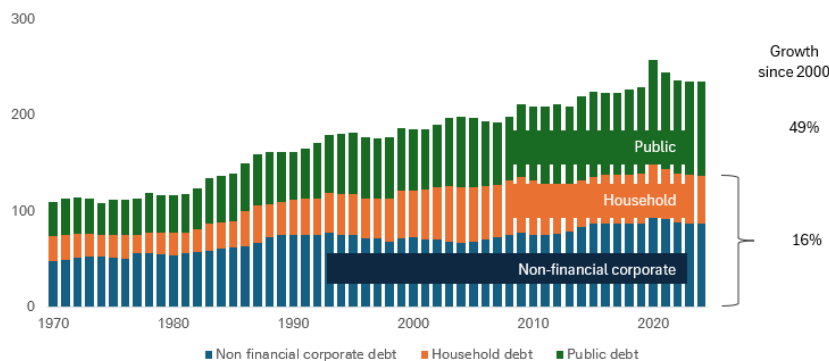
Why 'flat on the week' is the wrong takeaway

An index that round-trips 45 percentage points in days is not the same thing, economically, as one that simply sits still. Leveraged and margined positions are typically closed out mechanically and irreversibly on the way down, regardless of what happens on the way back up. The P&L may net out at the index level; the wealth destruction for individual, often first-time, leveraged investors does not.

We believe the potential for household leverage to cause problems is underestimated because the aggregate figures look okay. When we show the following chart, clients gravitate to the rise in sovereign debt, not the plateau in household leverage.

Figure 1: Household debt has lost the spotlight to public debt since 2000

Global debt as a % of global GDP



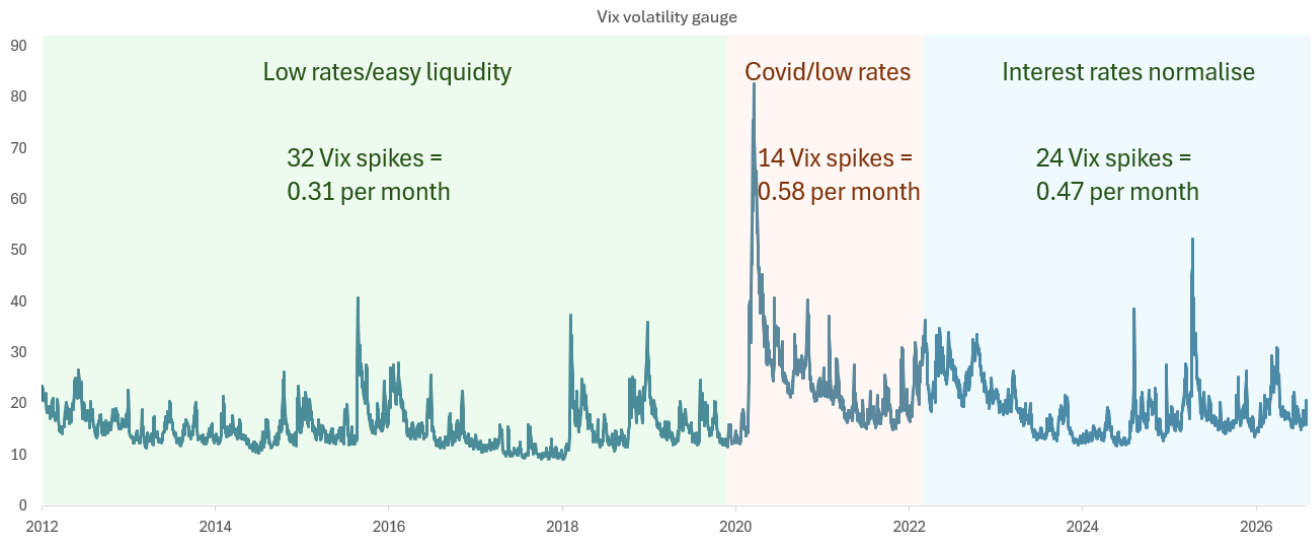
Source: IMF, Deutsche Bank Research

Small vol events are becoming more frequent

The following chart is created using our own judgement. But it shows that, by our reckoning, the number of spikes in the Vix since 2022 has been materially higher over the last few years than during last decade. The last four years are an important comparison period as this is when interest rates began to normalise.



Figure 2: A greater frequency of ‘small vol’ events appears to be the norm, at least for now



Source: Factset, Deutsche Bank Research

Is the Fed trying to bring volatility back on purpose?

One explanation for why policymakers may be tolerating episodes like these comes from Rob Armstrong (a former colleague) [here](#). He advanced a theory about Fed chair Kevin Warsh that essentially says, his withdrawal of forward guidance is a deliberate attempt to reintroduce ‘uncertainty premia’ into markets that have used lower volatility to take on more leverage than they otherwise would. The idea is that “Too little everyday volatility ultimately leads to financial crises.”

The mechanism, on this reading, is that forward guidance suppresses everyday volatility – and stable prices and calm markets encourage more borrowing, both by governments and by market participants, than would occur under normal price discovery. Warsh cannot say this outright without spooking markets, hence the vaguer ‘referee’ rhetoric – but the policy actions, in Armstrong’s view, point the same way.

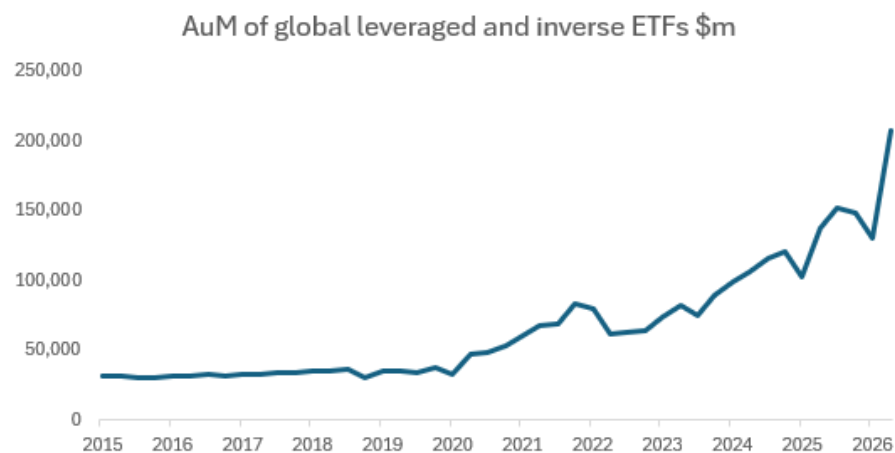
If policymakers genuinely believe more day-to-day volatility is healthy, investors should expect more ‘small vol’ episodes like Situational Awareness and the Kospi swing and should stop treating each one as isolated, ring-fenced events.

Leveraged ETFs and margin debt: Should we be afraid?

Leveraged and inverse ETFs, and the margin debt that often sits alongside them, are growing quickly across many major markets. They are a key mechanism that turns an ordinary correction into a cascade because they force mechanical, non-discretionary selling as prices move.



Figure 3: The semiconductor and broader tech rally has fuelled the popularity of leveraged ETFs



Source: Bloomberg Finance L.P., Deutsche Bank Research

In Korea specifically, the launch of single-stock leveraged ETFs tracking Samsung Electronics and SK Hynix in April turned two already-dominant, trillion-dollar-plus stocks into vehicles for retail leverage almost overnight. Retail investors accounted for a large proportion of holders. When the underlying stocks fell, those products amplified the move.

The risk is not confined to Korea. The same products are growing in the US and Europe. A market that is many times larger, and far more directly connected to consumer spending and credit availability, having its own 'Kospi moment' is a materially bigger problem for systemic risk than a niche hedge fund unwinding.

A footnote: why the name 'Situational Awareness' went viral

The fund's name borrowed from Leopold Aschenbrenner's 165-page 2024 essay, which argued AGI was plausible by 2027 and sketched a path from massive compute build-outs to an 'intelligence explosion' and eventual government-led consolidation of frontier AI ('The Project').

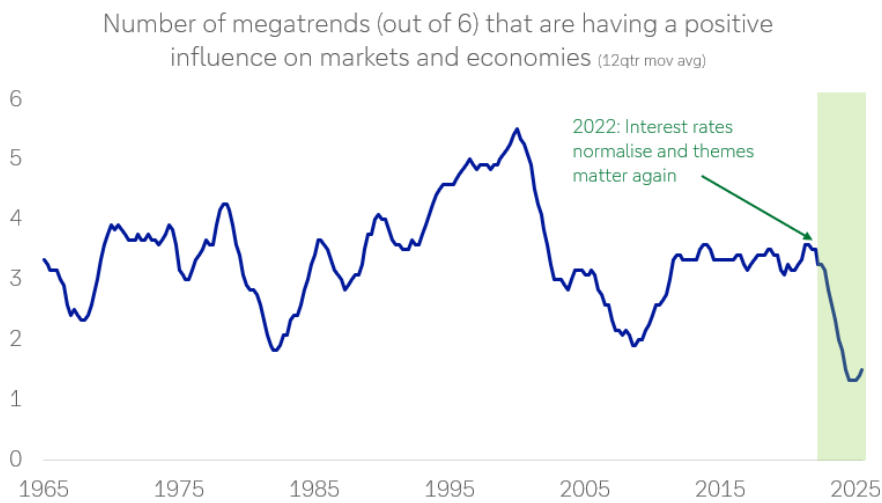
Part of a developing medium-term trend

Our megatrend framework tracks six long-run forces (technology, sovereign deficits, geopolitics & globalisation, domestic politics & social discontent, demography, and energy) and combines them into a single indicator of how supportive the backdrop is for markets and economies.

Overall, the number of megatrends working in favour of markets and economies has fallen to a level last seen only twice before – during the 1970s oil crises and around the 2008 financial crisis. Both were periods marked by exactly this kind of 'small vol' pattern: idiosyncratic-looking events that, in hindsight, were symptoms of a more fragile underlying system.



Figure 4: Prior periods of megatrends being this deeply negative have resulted in the greater frequency of volatility events



Source: Deutsche Bank Research

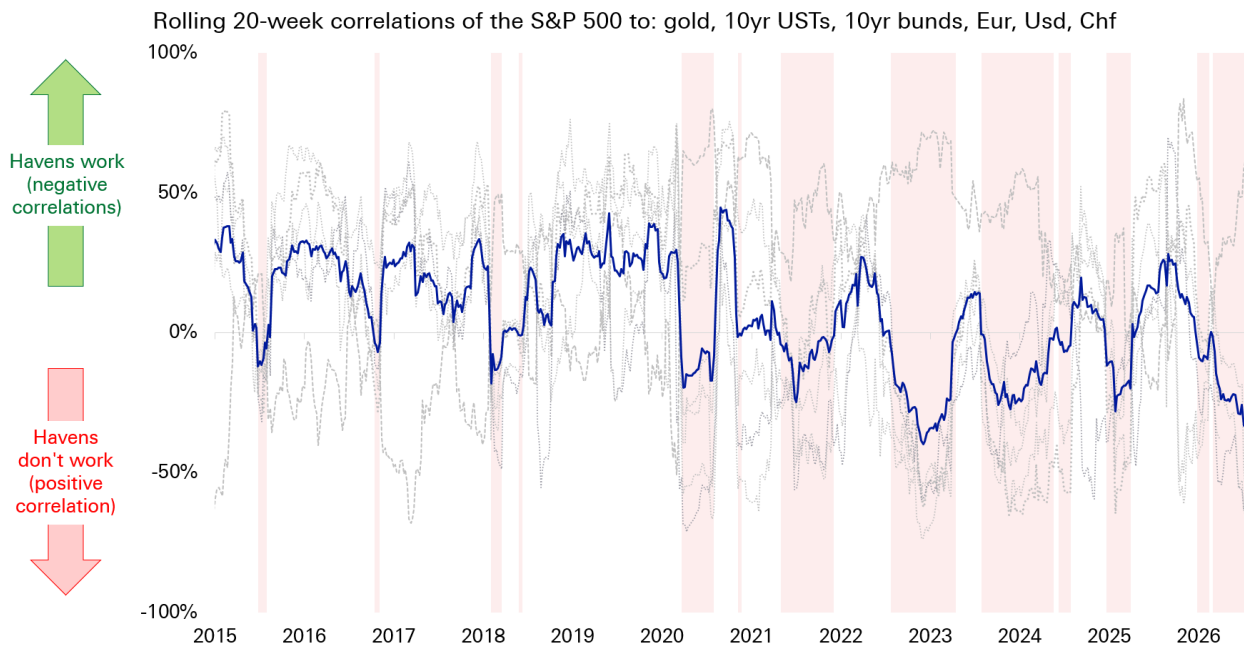
The single most important negative megatrend is sovereign deficits. Developed-market governments are refinancing and issuing new debt at rates well above those of the last decade. At the same time that our model flags unpredictable sovereign debt sell-offs, particularly those triggered by exogenous shocks and investor emotion, as a growing risk.

Havens are not there to catch you

Ordinarily, investors would look to traditional havens (gold, the dollar, the Swiss franc, the yen, US treasuries and bunds) to cushion volatility. However, our haven indicator shows that, since 2020, this combination of assets has failed to hedge risk (as proxied by the S&P 500) particularly during more shock events, including covid, the 2022 rate hikes, the 2025 tariff shock and the 2026 Iran war.



Figure 5: Traditional haven assets have not worked in the 2020s, particularly during shock events and periods of volatility



Source: Bloomberg Finance L.P., Deutsche Bank Research

If havens keep failing when they are needed most, the cushioning effect investors implicitly rely on to keep 'small vol' small is weaker than it used to be. That raises the odds that future 'small vol' episodes may cascade into something more.

What if this happened in the US or Europe?

It is tempting to treat the Kospi swing as a Korea-specific story — a function of retail culture, single-stock products and two dominant, correlated semiconductor names. But if a similar episode plays out in US or European, the results could be systemic.

The transmission mechanism would not be limited to portfolio losses. It would run through consumer spending, as households with margin-called leveraged positions cut back; through credit, as brokers and lenders reassess collateral and risk; and through confidence, as retail participation reverses. A leveraged retail shock in the US would be a materially larger systemic risk than a single hedge fund unwind.

Another example

Silicon Valley Bank is another example. On paper, SVB's failure should have been a contained, idiosyncratic event — a mid-sized bank with a concentrated depositor base and a duration mismatch. In practice, it triggered enough of a confidence panic that policymakers felt compelled to effectively guarantee all deposits across the banking system, a far larger intervention than the initial problem seemed to warrant.

The lesson is not that every small event becomes systemic but the ones that do are rarely obvious in advance. And when there is an increased frequency of these



events in a market with more leverage, less reliable havens, and policymakers newly tolerant of everyday volatility, there is an increased risk that 'small vol' events may catalyse larger problems.

Conclusion: are we situationally aware?

The market's instinct is to file Situational Awareness and the Kospi swing under 'noise' because neither, so far, has cascaded into a systemic event. That instinct may be dangerous, particularly if policymakers are willing to tolerate more everyday volatility.

Given our megatrend model indicates that the supportive backdrop for markets is near a multi-decade low, 'small vol' episodes can potentially be used as an indicator of how stress in the financial system is quietly building up.

There are plenty of other parts of the market where 'small vol' events may catalyse problems in areas that were built up with leverage during the periods of easy money. Just two examples are:

- Private equity: a large amount of TMT portfolio companies bought during the era of near-zero rates remain on PE firms' books, held longer than usual as exit markets stayed shut. A reckoning may still be ahead.
- Corporates: many listed companies still carry underperforming divisions bought during debt-fuelled M&A booms last decade. As a result, they have lower asset turnover metrics, a legacy of a decade in which cheap financing incentivised growth in revenue over earnings.

Appendix: Why Situational Awareness went viral

The original essay by Leopold Aschenbrenner in 2024 was 165 pages long. Below is a bullet point summary (generated by AI).

The Path to AGI by 2027: The document argues that Artificial General Intelligence (AGI) is strikingly plausible by 2027. This prediction is based on extrapolating "Order of Magnitude" (OOM) trendlines in three areas: physical compute scaling, algorithmic efficiencies, and "unhobbling" gains (transforming models from chatbots into active agents).

Massive Industrial Mobilisation: AI is shifting from a software boom to a massive industrial process. Leading AI labs will build training clusters costing \$100s of billions by 2028 and over \$1 trillion by 2030. By the end of the decade, these clusters could require power equivalent to 20% of total U.S. electricity production.

The Intelligence Explosion: Once AGI is achieved, it will likely be used to automate AI research itself. Because digital researchers can run in millions of copies and at much higher speeds than humans, a decade of algorithmic progress could be compressed into a single year, leading rapidly from AGI to superintelligence.

The Security Crisis: Current security at leading AI labs is described as "Swiss cheese," leaving key algorithmic secrets and model weights vulnerable to state-



actor theft, particularly by the CCP. The author warns that leaking these secrets in the next 12-24 months would be a catastrophic national security failure.

Superalignment Challenges: Reliably controlling AI systems that are vastly smarter than humans is an unsolved technical problem. Current alignment techniques (like Reinforcement Learning from Human Feedback) will break down because humans cannot supervise behaviour they do not understand. Solving this will require using AI to help automate alignment research.

"The Project" (U.S. Government Intervention): Given the decisive military and economic advantage of superintelligence, private startups cannot handle it alone. By 2027 or 2028, the U.S. government is expected to initiate a nationalized AGI effort—referred to as "The Project"—to consolidate resources, secure model weights, and ensure the free world prevails over authoritarian powers.

Geopolitical Stakes: Superintelligence will provide a military edge comparable to nuclear weapons. A lead of even a few months could be decisive, making a healthy lead for democratic allies essential for safety and global stability.



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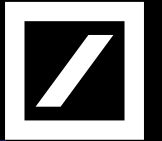
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This Month in Geopolitics: August 2026

Key events, themes and trends to watch in the month ahead

August 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in August 2026

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



In the spotlight for August 2026: US-Iran peace talks and US trade policy



US-Iran war (all of August)

Expect more volatility – but also efforts to resume dialogue – during August. Tensions escalated significantly in July, with renewed military strikes and the US reimposing sanctions on Iranian oil sales. Both sides appear unwilling to concede for fear of losing ground. While President Trump has suggested that the ceasefire is over, **we retain our assessment that both parties are incentivized to find a near-term resolution** as kinetic attacks will not drive a resolution. That said, we should prepare for more volatility in the coming months as Iran is unwilling to cede control of the Strait of Hormuz and neither side is prepared to lose face. The Yemen-based **Houthi blockade of Saudi Arabia**, announced on 20 July, further heightens risks by threatening crude oil flows via Saudi Arabia's Yanbu on the Red Sea (see map), a terminal which handles about 4 million barrels of crude oil and condensate exports per day.

The persistent security risks will likely continue to drive the Gulf states to reduce their vulnerability to the Strait of Hormuz. Shipping levels through the Strait will periodically be impacted by Iranian threats to navigation and political considerations. Even once dialogue resumes and fighting subsides, **traffic will likely take time to recover** given the challenges of de-mining, a lack of trust between the US and Iran, and Iranian threats. Infrastructure repair to ports, terminals and energy infrastructure, as well as the provision of security guarantees are other factors to watch. Full operational normalization, involving infrastructure repairs, [is not expected](#) until 2027 and beyond.

US trade policy (all of August)

When the US 10% global Section 122 tariffs expired **on 24 July**, the Trump administration moved quickly to fill the gap with Section 301 tariffs of 10-12.5% on more than 60 countries. In August, we should expect the Trump administration to **keep building its replacement “permanent tariff framework”** using Section 301 and 232 authorities. The USTR's Section 301 investigations into “structural excess capacity” in 16 manufacturing-heavy economies remains ongoing and could yield new tariffs in the coming weeks. The Trump administration is also pursuing several Section 232 national security investigations with scheduled tariff effective dates of 31 July and 29 Sept. This should keep pressure on countries to strike bilateral deals favorable to the US, spanning trade, investment, industrial and technology concessions.

To watch will be movement on unresolved country-specific issues, e.g., India's tariff talks, the EU's proposed 30% tariff, in addition to continued legal uncertainty as court appeals over the tariffs' legal basis play out. This will set a complex backdrop to the G20 finance ministers' meeting where trade protectionism is already shaping up to be a point of friction.

Key regional ports in the Persian Gulf



Source: Deutsche Bank. For more on the impact of the Iran conflict, see [Twenty Miles That Shook the World - Deutsche Bank Research Institute](#)

Key events, issues and themes to watch in August 2026

Listed in chronological order



US-Mexico-Canada trade talks (all of August)

The renewal talks are expected to be challenging and potentially protracted. 1 July marked the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process whereby the three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode. The US has declined to confirm the renewal of USMCA "in its current form".

In August, we expect further US-Mexico-Canada negotiations focused on resolving outstanding issues (e.g., automotive rules of origin, steel/aluminum, economic security, and issues around agriculture, labour and environment). At the G7 summit, Canada and the US agreed to a 30-day countdown for a new cross-border economic/security agreement, which was extended to 1 August.

In any case, Mexico and Canada are both subject to a 10% Section 301 tariff that took effect on 24 July. In addition, the US 50% tariffs on a range of Canadian goods announced on 20 July, and additional Section 232 investigations (lumber, copper, semiconductors, pharmaceuticals, trucks, critical minerals, etc.) could complicate talks further as Mexico and Canada are top exporters in several of these sectors. The Trump administration is also using the review to push non-trade demands on migration, drug trafficking and defence cooperation.

First weeks of new UK Prime Minister (all of August)

Andy Burnham became the UK's 7th Prime Minister since 2016 on 20 July. He enters office facing a sluggish economy, cost-of-living pressures, and a massive welfare bill. He has pledged this moment as a "circuit-breaker" for Britain and has quickly set out to focus on concrete cost of living measures. In August, we can expect Mr. Burnham to be busy with the early scaffolding for his "No. 10 North" project focused on devolution of power. Mr. Burnham has signaled his priorities will be cost-of-living issues, devolution, public control of essentials, social commitments and longer-term structural reform. UK parliament is typically in summer recess through August, which limits legislative action but gives Mr. Burnham room to set the tone for his premiership with an intention to move quickly on key commitments in the autumn.



Prime Minister Andy Burnham was sworn in on 20 July. Source: [House of Commons](#)

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China-US talks ahead of next Xi-Trump summit (all of August)

Bilateral dialogue to tee up deliverables for the 24 September summit will be a key watchpoint during August. Recent developments – including US allegations of [election interference](#) and warnings over [Chinese AI models](#) – reinforce our view that relations remain structurally competitive despite the recent stabilisation in ties. Investors should anticipate more, not fewer, bilateral economic restrictions over the medium term.

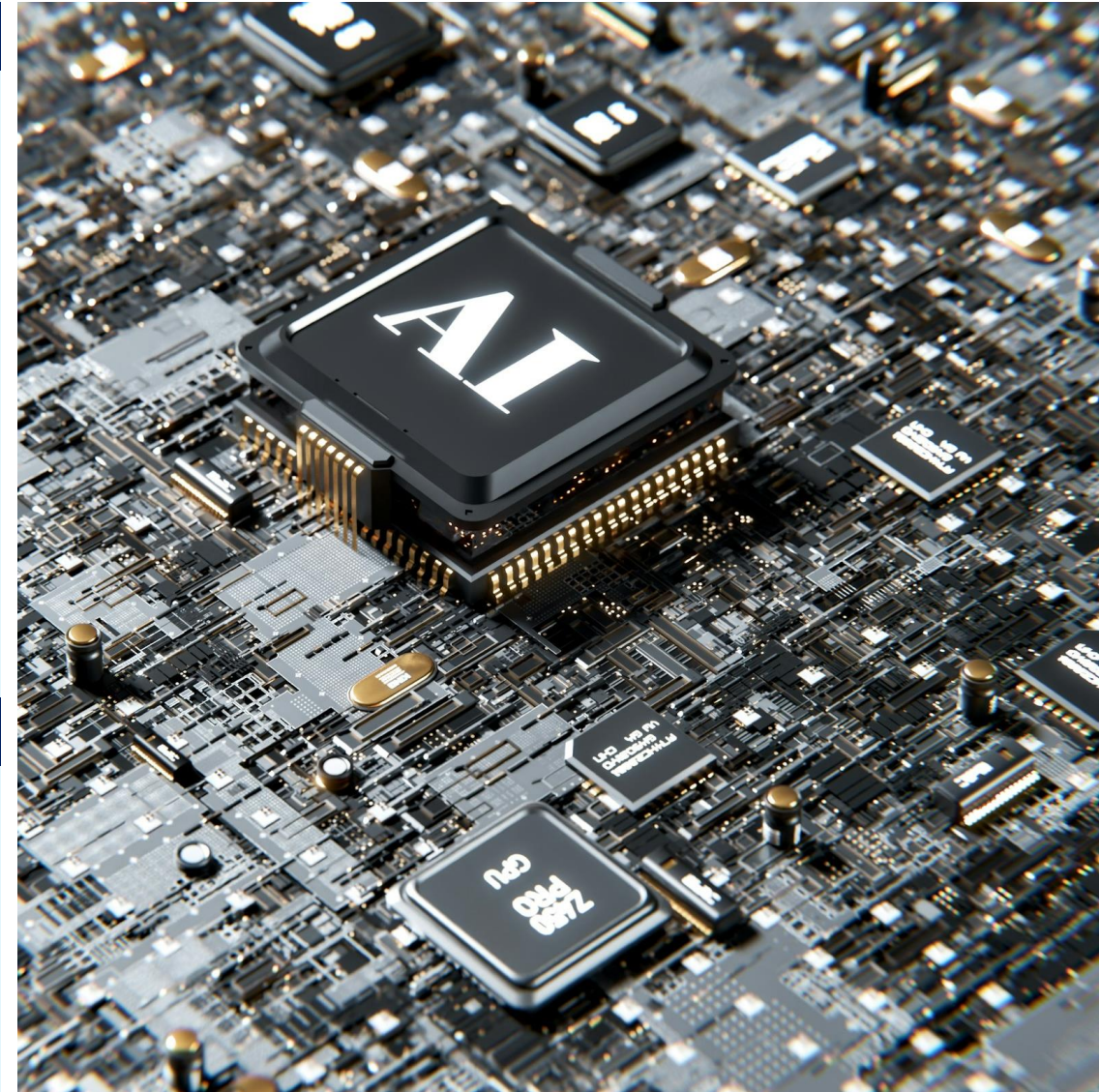
Nevertheless, there is currently no evidence to suggest these dynamics are likely to delay the next Xi-Trump summit, which will likely go ahead as planned on 24 September. During August, investors should watch for any sign of progress on talks on the Board of Trade (intended to boost non-sensitive trade), Board of Investment (to facilitate investment in non-sensitive areas) and an extension of the Busan agreement – which delayed a range of new economic restrictions until November 2026.

Both sides could also strengthen commitments around AI dialogue, with first talks [reportedly](#) expected to happen ahead of the summit. Dialogue on AI was a key deliverable from the May Xi-Trump summit, and both sides could seek to address persistent tensions around the technology, although a major breakthrough appears unlikely.

China's leaders gather for summer retreat in Beidaihe (early August)

The retreat provides an opportunity for Party leaders to exchange views on major policy issues. This year's meeting could potentially cover topics ranging from the conflict in Iran, relations with the EU and US, the implementation of the 15th Five-Year Plan (2026-2031) and early deliberations around the next Party Congress in late 2027.

The retreat primarily serves as an opportunity for internal alignment and consensus building on key domestic and external policy issues. There are no official readouts, but subsequent state media framing, leadership signals, and Party meetings in the autumn could offer clues on the direction of policy debate.



Source: [Igor Omilaeu](#) on [Unsplash](#)

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Abelardo de la Espriella inaugurated as Colombia's next president (7 August)

The president's early agenda is expected to include significant policy shifts, including a planned urban security coordination mechanism for major cities; a more pro-US and pro-Israel foreign policy stance; and a review of Colombia's diplomatic posture toward Cuba, Nicaragua, Venezuela and multilateral bodies. Key watchpoints over the next 2-3 months will include cabinet formation and congressional coalition-building. The incoming administration has also vowed to disrupt narcotics production and trafficking, and counter violence by armed groups operating in parts of the country – including in a bid to attract more foreign investment.

Kazakhstan legislative elections (23 August)

Kazakhstan will elect all 145 deputies to the newly-created Kurultai. This will be the first election to the new unicameral legislature established after March 2026 constitutional reforms replaced the old bicameral parliament. These elections are unlikely to materially impact governance in the country with President Tokayev remaining firmly in control of the political system. Kazakhstan's constitutional court ruled that president Tokayev may seek another term under the new 2026 constitution, effectively resetting his term count despite having been elected in 2022 to what was billed as a single, non-renewable seven-year term ending 2029. While the election itself is unlikely to produce policy surprises, it will be an indicator of the government's ability to manage the political transition under the new constitutional framework and maintain domestic stability.

Iceland referendum on resuming EU accession talks (29 August)

Iceland will hold a referendum asking voters whether the country should resume EU accession negotiations. Iceland had applied to join the EU in 2009, but suspended accession talks in 2013 after a change of government and later withdrew the application.

The current coalition government pledged to finally let voters decide, and accelerated the timeline partly due to shifting geopolitical dynamics, including US tariff hikes and Trump administration statements on acquiring Greenland. Polling is tight with surveys showing 52% in favor with 28% opposed. A vote to resume negotiations would not commit Iceland to EU membership but would mark a significant shift in the country's strategic orientation.



Abelardo de la Espriella will be inaugurated as Colombia's next president on 7 August. Source: [Wikimedia Commons](#)

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G20 Meeting of Finance Ministers and Central Bank Governors (31 Aug to 1 Sep)

This will be a key checkpoint in the US' G20 presidency, where Washington sets the agenda for global economic policy coordination among the world's largest economies. The US has identified as its priorities: modernizing financial regulation, strengthening understanding of excessive global imbalances, enhancing debt transparency and facilitating debt restructuring, endorsing digital assets and improving cross-border payments. Treasury Secretary Scott Bessent has sought a “streamlined, results-oriented” agenda.

The first G20 finance ministers meeting under the US presidency in April failed to produce a joint communiqué or chair statement and no joint press conference was held at the end. The current crisis in the Middle East amongst other geopolitical developments may crowd out purely technical finance discussions.

Washington's tighter focus may clash with other members who still want to discuss debt relief for developing countries, climate finance and inequality. With a US leadership openly skeptical of the G20's broader mission, to watch will be how much substantive coordination is feasible.



Source: [Giorgio Trovato](#) on [Unsplash](#)

Appendix 1



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